FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Pre-eclampsia & Eclampsia

Tutorial | Week 4 · Hypertension, twins & the pelvic floor | Obstetrics | Dr Nanzira

Pre-eclampsia is a placenta-driven, multisystem endothelial disease of pregnancy (hypertension ≥140/90 after 20 weeks plus proteinuria or organ/placental dysfunction). It kills through stroke, eclampsia, HELLP, pulmonary oedema and abruption; $MgSO_4$, prompt control of severe BP and timely delivery are what save lives.

Must know before the ward round

1. Pre-eclampsia = new BP ≥140/90 after 20 weeks + proteinuria (PCR ≥30 mg/mmol or dipstick ≥2+) OR maternal organ dysfunction OR uteroplacental dysfunction (FGR).
2. Proteinuria is not required for the diagnosis, and the amount of proteinuria does not grade severity - platelets, liver, kidney, brain, lungs and BP do.
3. Severe hypertension (≥160/110, confirmed within 15 min) is an emergency - treat within 30–60 minutes (nifedipine IR 10 mg oral, hydralazine 5 mg IV, or labetalol 20 mg IV); aim for 140–150/90–100.
4. $MgSO_4$ Pritchard (Uganda) - 4 g of 20% IV over 5–10 min + 10 g of 50% IM (5 g each buttock); then 5 g IM every 4 h in alternate buttocks for 24 h after delivery or last fit.
5. Before every $MgSO_4$ dose - RR ≥16/min, patellar reflexes present, urine ≥30 mL/h (≥100 mL in 4 h). Antidote - calcium gluconate 1 g (10 mL of 10%) IV over 10 min.
6. Any fit after 20 weeks is eclampsia until proven otherwise - left lateral, airway, oxygen, $MgSO_4$ loading dose, then check glucose and mRDT.
7. HELLP = Haemolysis (LDH ≥600 IU/L), Elevated Liver enzymes (AST ≥70 IU/L), Low Platelets (<100 × 10^9/L).
8. Severe-feature bloods - platelets <100, AST/ALT >70 IU/L, creatinine >90 µmol/L; bedside clotting test with no clot at 7 min = coagulopathy.
9. Delivery is the only cure. Eclampsia - deliver within 12 h of stabilisation; severe PE ≥34 weeks - within 24 h; pre-eclampsia without severe features - deliver at 37 weeks.
10. Spinal anaesthesia with platelets ≥75–80 × 10^9/L; transfuse platelets if <50 before caesarean or <20 at any time.
11. Aspirin 75–150 mg at night from 12 weeks (before 16–20 weeks) to 36 weeks for high-risk women; calcium 1.5–2 g/day where dietary intake is low.
12. Avoid ergometrine in hypertensive women; restrict IV fluids to about 80 mL/h (1 mL/kg/h) - pulmonary oedema kills.

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1. Definition

- **Pre-eclampsia** is new hypertension after 20 weeks with evidence that the disease has spread beyond the blood vessels: protein in the urine, damage to kidney, liver, brain, blood or lungs, or a struggling placenta (fetal growth restriction).
- It starts in the **placenta** (poor remodelling of the spiral arteries in early pregnancy) and ends in the **mother's endothelium** (the lining of every blood vessel). That is why it affects every organ, and why **delivering the placenta is the only cure**.
- **Eclampsia** is a generalised tonic-clonic seizure in a woman with pre-eclampsia (sometimes the first sign; up to a third of fits occur postpartum).

Hypertension in pregnancy = systolic BP ≥140 mmHg and/or diastolic BP ≥90 mmHg, on two readings (ideally at least 4 hours apart for non-severe values; within 15 minutes for severe values). **Severe hypertension** = ≥160 systolic and/or ≥110 diastolic.

What counts as proteinuria?

Test	Threshold
Urine dipstick	≥2+ is significant; 1+ needs confirming (PCR or 24-h urine)
Spot urine protein:creatinine ratio (PCR)	≥30 mg/mmol (≥0.3 mg/mg)
24-hour urine protein	≥300 mg/24 h

Three drugs and one decision save lives: **$MgSO_4$** (prevents and treats fits), a **rapid-acting antihypertensive** (prevents stroke), and **timely delivery**.

> CLINICAL PEARL
> The one thing never to forget: **a BP of ≥160/110 is an emergency. Treat it now, give $MgSO_4$, and plan delivery.** Do not wait for proteinuria, and do not be reassured by "no symptoms".

2. Epidemiology

- Pre-eclampsia complicates roughly **3–8% of pregnancies** worldwide; hypertensive disorders together complicate about 10%.
- Worldwide, hypertensive disorders cause about **14% of maternal deaths**. In low-income settings, eclampsia is far commoner (up to 10 times) than in high-income countries, because women present late, ANC BP checks are missed, and $MgSO_4$ and antihypertensives are not always on the shelf.
- In **Uganda** (maternal mortality ratio about 189 per 100,000 live births, UDHS 2022), hypertensive disorders are the **second leading direct cause of maternal death after PPH** in MPDSR reports. Most of those deaths are from **uncontrolled severe hypertension (stroke), eclampsia and pulmonary oedema**, which are largely preventable. Perinatal losses come from abruption, stillbirth, FGR and iatrogenic preterm birth.
- A national evaluation of emergency care for severe pre-eclampsia in Uganda found wide variation between facilities. Nifedipine was on the shelf in most HC III+ facilities (about 93%), but **IV/oral labetalol was essentially unavailable**, and the facilities that had protocols on the wall and regular drills delivered safer, faster care.
- A cohort at **Kawempe National Referral Hospital** confirmed that the **Pritchard (IM) regimen** is the standard of care in Uganda, and that giving IM maintenance doses on time is a real challenge on congested wards (missed doses are common).

UGANDA CONTEXT

Facility realities you will meet: BP machines that are broken or have the wrong cuff size; urine dipsticks out of stock; $MgSO_4$ available but calcium gluconate missing; only one midwife for 30 mothers, so 4-hourly IM doses are missed; no ICU bed; and referral delays because of transport. Your job as a clerk is to **measure the BP properly, dip the urine, check the $MgSO_4$ chart and reflexes, keep the fluid balance chart** and escalate early.

3. Aetiology and Risk Factors

The cause is abnormal placentation (section 4); immune maladaptation to paternal antigens is part of why **first pregnancies and new partners** carry higher risk.

High risk (any ONE → aspirin)	Moderate risk (≥2 → aspirin)
Pre-eclampsia in a previous pregnancy	First pregnancy
Chronic hypertension	Age ≥40 years
Chronic kidney disease	Pregnancy interval >10 years
Autoimmune disease (SLE, antiphospholipid syndrome)	BMI ≥35 kg/m² at booking
Type 1 or type 2 diabetes	Family history of pre-eclampsia (mother or sister)
	Multiple pregnancy (large placental mass)

Other associations: molar pregnancy (pre-eclampsia before 20 weeks), IVF, new partner, sickle cell disease.

4. Pathophysiology

Understanding the mechanism makes every management step logical.

Stage 1: abnormal placentation (first half of pregnancy, silent)

1. In normal pregnancy, **extravillous trophoblast** invades the decidua and the inner third of the myometrium and **remodels the maternal spiral arteries**. It destroys their muscular, elastic walls, turning them into wide, floppy, low-resistance channels that deliver a high volume of blood at low pressure to the intervillous space.
2. In pre-eclampsia this remodelling is **shallow and incomplete**. The myometrial segments stay narrow and muscular, so they can still constrict.
3. Result: **placental hypoperfusion**, intermittent ischaemia-reperfusion, and **oxidative stress** in the syncytiotrophoblast.

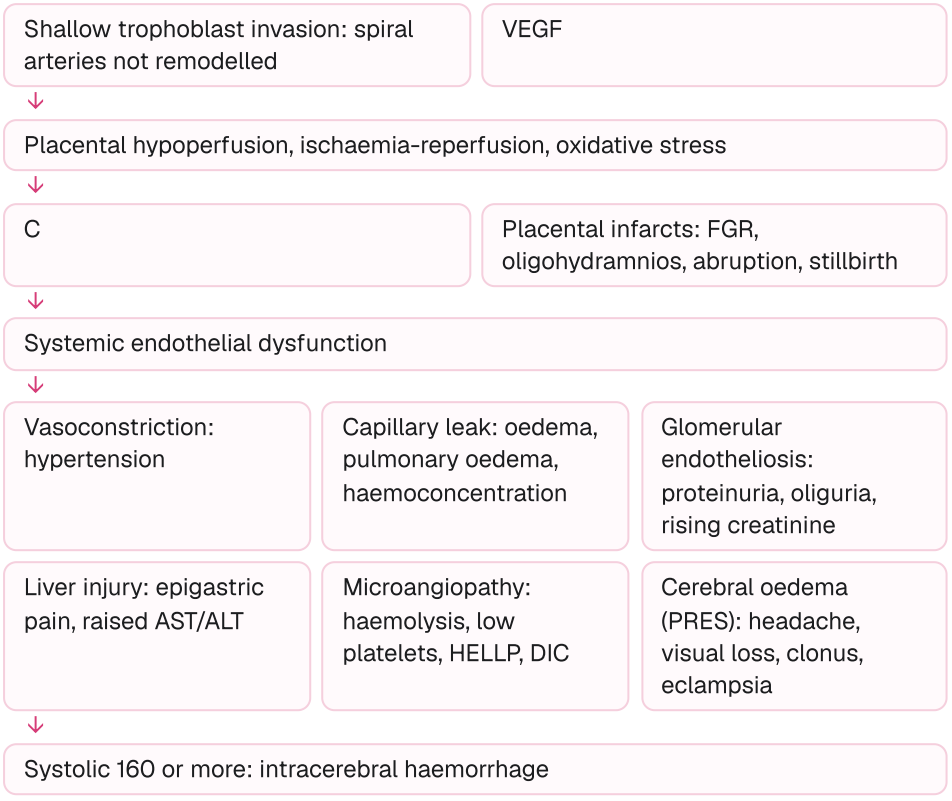
Stage 2: maternal syndrome (after 20 weeks)

4. The stressed placenta releases **anti-angiogenic factors**, chiefly **soluble fms-like tyrosine kinase-1 (sFlt-1)** and **soluble endoglin**. These mop up the pro-angiogenic factors **VEGF** and **PlGF** (placental growth factor) that keep the endothelium healthy.
5. **Widespread endothelial dysfunction** follows:

Endothelial change	Clinical consequence	Why it matters for management
↓ Nitric oxide and prostacyclin, ↑ thromboxane, ↑ sensitivity to angiotensin II	**Vasoconstriction → hypertension**	Antihypertensives treat the number, not the disease

Endothelial change	Clinical consequence	Why it matters for management
Leaky capillaries, low plasma oncotic pressure	**Oedema, pulmonary oedema**, yet the circulation is **volume-depleted** (haemoconcentration)	Restrict IV fluids (about 80 mL/h). Do not "fill up" for oliguria
Glomerular endotheliosis (swollen endothelial cells block the capillary lumen)	**Proteinuria**, ↓ GFR, rising uric acid and creatinine	Monitor urine output; adjust $MgSO_4$ (renally excreted)
Periportal haemorrhage and necrosis, liver capsule stretch	**Epigastric/RUQ pain**, ↑ AST/ALT, rarely **subcapsular haematoma/rupture**	Epigastric pain is a severe feature, not "gastritis"
Microangiopathy (fibrin on damaged endothelium)	**Haemolysis** (schistocytes, ↑ LDH) and **platelet consumption** → HELLP, DIC	Check platelets before spinal anaesthesia; watch for bleeding
Loss of cerebral autoregulation, vasogenic oedema (PRES)	**Headache, visual disturbance, hyperreflexia, eclampsia**, haemorrhagic stroke when systolic ≥160	Control systolic BP; $MgSO_4$ stabilises neurons
Placental infarcts, spiral artery atherosis	**FGR, oligohydramnios, abruption, stillbirth**	Fetal surveillance; the fetus may dictate delivery

6. **Why delivery cures it:** the source of sFlt-1 is the placenta. After delivery, levels fall over days, which is why the disease can **worsen in the first 48–72 hours postpartum** before it resolves.



Feature links

- Vasoconstriction → raised systemic resistance → hypertension; systolic ≥160 → loss of cerebral autoregulation → haemorrhagic stroke.
- Capillary leak and low oncotic pressure → sudden facial and hand oedema, rapid weight gain → pulmonary oedema (crackles, SpO_2 <95%), especially after IV fluids or postpartum.
- Capillary leak → haemoconcentration → high Hb/PCV despite oedema.

- Glomerular endotheliosis → proteinuria, falling GFR → oliguria, rising creatinine and uric acid → magnesium accumulates → toxicity.
- Hepatocellular injury and capsule stretch → epigastric/RUQ pain, nausea, vomiting, raised AST/ALT → subcapsular haematoma (shoulder-tip pain, shock).
- Microangiopathic haemolysis → schistocytes, raised LDH and unconjugated bilirubin, falling Hb, jaundice.
- Platelet consumption → thrombocytopenia → bruising, oozing, spinal haematoma risk.
- Cerebral vasogenic oedema → severe headache, blurring, flashing lights, scotomata, cortical blindness → hyperreflexia, ankle clonus → seizure.
- Retinal vasospasm or detachment → sudden visual loss.
- Placental insufficiency → fetal growth restriction, oligohydramnios, reduced fetal movements, abnormal Doppler.
- Damaged placental bed → abruption → vaginal bleeding, hard tender uterus, DIC, IUFD.
- Magnesium excess (renal failure) → loss of patellar reflexes → respiratory depression → cardiac arrest.

Why $MgSO_4$ works. $MgSO_4$ is **not** mainly an antihypertensive. It is a **calcium antagonist** that causes cerebral vasodilatation (relieving vasospasm), an **NMDA-receptor antagonist** that raises the seizure threshold, and it reduces blood-brain-barrier permeability (less cerebral oedema). The Magpie Trial showed it **halves the risk of eclampsia**, and the Collaborative Eclampsia Trial showed it is **better than diazepam or phenytoin** at preventing recurrent fits and maternal death.

Why $MgSO_4$ can be toxic. Magnesium is excreted **only by the kidneys**. In pre-eclampsia the kidneys are often failing (oliguria), so magnesium accumulates. Toxicity follows a predictable ladder: **reflexes go first, then breathing, then the heart**. That is why we check **reflexes, respiratory rate and urine output** before every dose.

Serum Mg^{2+}	Effect
0.7–1.0 mmol/L	Normal
2–3.5 mmol/L (4.8–8.4 mg/dL)	**Therapeutic range**
4–5 mmol/L (about 10–12 mg/dL)	**Loss of patellar reflexes**
6–7.5 mmol/L (about 15–17 mg/dL)	**Respiratory depression / paralysis**
>12 mmol/L (>30 mg/dL)	**Cardiac arrest**

5. Classification / Staging

Category	Definition	Key point
Chronic (pre-existing) hypertension	BP ≥140/90 before pregnancy or before 20 weeks, or persisting beyond 12 weeks postpartum	Look for a cause (renal, etc.). Risk of superimposed pre-eclampsia is about 25%
Gestational hypertension	New BP ≥140/90 at or after 20 weeks, **without** proteinuria or other features of pre-eclampsia	About a quarter progress to pre-eclampsia, so keep checking urine and bloods
Pre-eclampsia	Gestational hypertension **plus** ≥1 new feature: proteinuria, maternal organ dysfunction or uteroplacental dysfunction	ISSHP 2021 definition; proteinuria is **not** required
Pre-eclampsia superimposed on chronic hypertension	Woman with chronic hypertension who develops new proteinuria, organ dysfunction or FGR	Sudden worsening of BP control is a clue
Eclampsia	New generalised tonic-clonic seizures in a woman with pre-eclampsia, with no other cause	Can occur antepartum, intrapartum or postpartum

Category	Definition	Key point
HELLP syndrome	Haemolysis, Elevated Liver enzymes, Low Platelets	A severe variant of pre-eclampsia. BP may be only mildly raised and proteinuria may be absent
White-coat hypertension	High BP in clinic, normal at home/ambulatory	Still carries some increased risk

Features of severe disease ("pre-eclampsia with severe features"). Any **one** of the following makes it severe:

System	Severe feature
Blood pressure	≥160 systolic or ≥110 diastolic (confirmed within 15 minutes)
Neurological	Severe persistent headache, visual disturbance (blurring, flashing lights, scotomata, cortical blindness), altered consciousness, hyperreflexia with clonus, eclampsia, stroke
Respiratory	Pulmonary oedema (breathlessness, crackles, SpO_2 <95%)
Hepatic	Epigastric or right upper quadrant pain; transaminases ≥2× upper limit of normal (AST or ALT >70 IU/L)
Haematological	Platelets <100 × 10^9/L; haemolysis; DIC
Renal	Creatinine >90–97 µmol/L (>1.1 mg/dL) or doubling; oliguria <30 mL/h (<500 mL/24 h)
Fetoplacental	Fetal growth restriction, abnormal umbilical artery Doppler, abruption, stillbirth

> NOTE
> Older Ugandan teaching (and some UCG tables) grade pre-eclampsia as "mild/moderate" versus "severe" using diastolic BP and the number of pluses of proteinuria. Modern guidance (ISSHP, ACOG, NICE) has dropped **amount of proteinuria** as a marker of severity. Heavy proteinuria still matters (it predicts earlier delivery), but a woman with 1+ protein and a platelet count of 60 has **severe** disease.

HELLP thresholds (Tennessee): LDH ≥600 IU/L, AST ≥70 IU/L, platelets <100 × 10^9/L.

> NOTE
> Some units also use the **Mississippi classification** of HELLP by platelet nadir (class 1 ≤50, class 2 >50–100, class 3 >100–150 × 10^9/L). The lower the platelets, the higher the risk of bleeding and DIC. Thresholds differ slightly between classifications; the Tennessee thresholds are the ones in these notes.

6. Clinical Features

History

Ask	Why it matters (what answer changes the plan)
Gestational age (LNMP, earliest scan, fundal height at booking)	<20 weeks → think **molar pregnancy** or chronic hypertension. <34 weeks → **steroids**; <32 weeks → $MgSO_4$ also gives **fetal neuroprotection**. ≥37 weeks → deliver
Headache (where, severity, relieved by paracetamol?)	Severe frontal/occipital headache not relieved by simple analgesia = **cerebral irritation → imminent eclampsia**; a **severe feature** → $MgSO_4$ and plan delivery

Ask	Why it matters (what answer changes the plan)
Visual disturbance (blurring, flashing lights, spots/scotomata, loss of vision)	Severe feature (occipital cortex oedema/PRES, retinal vasospasm). Sudden blindness = cortical blindness or retinal detachment
Epigastric / RUQ pain, nausea, vomiting	Liver capsule stretch → **HELLP** or subcapsular haematoma. Often misdiagnosed as "PUD/gastritis". Changes the bloods you send (LFT, LDH, film) and urgency
Breathlessness, orthopnoea, cough with frothy sputum	Pulmonary oedema (especially after IV fluids or postpartum) → stop IV fluids, sit up, oxygen, furosemide
Fetal movements	Reduced → FGR/hypoxia, abruption or IUFD; changes timing and mode of delivery
Vaginal bleeding, constant abdominal pain, hard uterus	**Placental abruption** (common with pre-eclampsia) → coagulopathy risk; deliver
Urine output, swelling of face/hands	Oliguria = renal involvement. Sudden facial oedema is a warning sign (though leg oedema alone is common and non-specific)
Any fit, loss of consciousness; fever, rigors, neck stiffness	Eclampsia vs **cerebral malaria**, meningitis, epilepsy
Drugs already given (at referring facility: $MgSO_4$ time and dose, hydralazine, nifedipine, diazepam, fluids)	Avoid double-loading $MgSO_4$; know when the next dose is due; count fluids already given
Past obstetric history: previous PE, eclampsia, HELLP, abruption, stillbirth, early delivery	Recurrence risk; shows how severe it was before
Chronic hypertension, kidney disease, diabetes, SLE, sickle cell, HIV; medications	Superimposed disease; drug interactions; methyldopa already started?
Asthma, heart disease	Avoid **labetalol**; caution with fluids
ANC: attended? booking BP, urine results, aspirin/calcium given	Booking BP ≥140/90 → chronic hypertension; explains late presentation; missed prevention (audit point)
Parity, previous CS, current contractions	Mode of delivery and likelihood of a quick vaginal birth
Where does she live, how far from a theatre?	Decides whether she can go home (almost never with severe disease in Uganda)

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination. In a fitting or drowsy woman, the **eclampsia algorithm** (section 9) comes first.

Introduction, consent and preparation

- Introduce yourself, confirm her identity and gestation, explain, and gain consent; chaperone for abdominal and vaginal examination.
- **Hand hygiene**.
- **Position and exposure**: seated or **left lateral at 45°** (avoid supine hypotension); expose the abdomen, keep her covered otherwise.

General inspection and vital signs

Sign	Mechanism	What it means
GCS, restlessness, confusion	Cerebral oedema, post-ictal, hypoglycaemia, malaria	Imminent or recent eclampsia; look for mimics
Pallor	Haemolysis, anaemia, abruption	HELLP; bleeding

Sign	Mechanism	What it means
Jaundice	Haemolysis, liver injury	HELLP, AFLP, malaria
Bruising, petechiae, oozing from cannula sites	Low platelets, DIC	Severe disease; no spinal
Oedema of face, hands, sacrum	Capillary leak	Sudden generalised oedema is more worrying than ankle oedema
Weight		For fluid calculation (1 mL/kg/h)

Blood pressure, measured properly: seated or left lateral at 45°, **correct cuff size** (a large cuff if mid-arm circumference >33 cm, otherwise the reading is falsely high), arm at heart level, palpate the systolic first, **Korotkoff V** (disappearance) for diastolic. Repeat after 15 minutes if severe. Record the arm and the position.

Pulse, **RR** (count before any $MgSO_4$ as a baseline), **SpO_2**, temperature (fever → malaria, sepsis, chorioamnionitis).

Hands, face and neck

- Hands: oedema, bruising, oozing from cannula sites; tremor or warmth (thyrotoxicosis in the differential).
- Face: facial oedema, jaundice, pallor.
- Eyes: **fundoscopy** if possible (papilloedema, haemorrhages).
- Neck: **JVP** (raised in fluid overload), neck stiffness (meningitis), goitre (thyrotoxicosis).

Chest

Basal crackles, raised JVP → pulmonary oedema, fluid overload.

Obstetric abdominal examination

- **Inspection**: size for dates, scars.
- **Palpation**: **epigastric/RUQ tenderness** (liver); uterine **tone and tenderness** (hard, tender = abruption).
- **Fundal height**: small for dates → FGR, oligohydramnios.
- **Lie and presentation**: breech or transverse → CS planning.
- **Engagement** (fifths palpable).
- **Auscultation of the fetal heart** for a full minute (Pinard or Doppler).

Neurological

Patellar reflexes (brisk, normal or absent: absent on $MgSO_4$ = toxicity; use the biceps reflex after spinal anaesthesia), **ankle clonus** (≥3 beats = cerebral irritability), focal deficits (stroke), neck stiffness.

Completing the examination

- **Vaginal examination** only when delivery is planned: **Bishop score** decides misoprostol/Foley versus ARM + oxytocin versus CS. Look for blood (abruption) and liquor colour.
- **Urine dipstick** for protein (every admission, every visit).
- Thank her, hand hygiene, summarise.

Document the key negatives: "no headache, no visual symptoms, no epigastric pain, no clonus, chest clear, uterus soft and non-tender, FHR 140 regular".

Red flags

RED FLAG

- BP ≥160/110 (even once, if confirmed within 15 minutes)
- Severe headache, visual disturbance, clonus → **imminent eclampsia**
- Epigastric/RUQ pain, vomiting → HELLP or liver haematoma
- Breathlessness, SpO_2 <95%, crackles → pulmonary oedema
- Urine output <30 mL/h (or <100 mL in 4 h)
- Platelets <100 × 10^9/L, rising creatinine, rising AST/ALT
- Vaginal bleeding with a tender, hard uterus → abruption
- Reduced or absent fetal movements, abnormal FHR
- Any seizure, drowsiness or confusion
- Loss of patellar reflexes or RR <16/min on $MgSO_4$ → **toxicity**

RED FLAG

Do not delay $MgSO_4$ or the antihypertensive to wait for lab results, a scan or a senior. Bloods are drawn **while** you treat.

Thinking like a clinician

- **"A fit in pregnancy after 20 weeks is eclampsia until proven otherwise."** Give $MgSO_4$ first, then look for malaria, hypoglycaemia and meningitis in parallel. Do not waste the golden hour on a lumbar puncture.
- **BP can be "only" 140/90 in eclampsia and HELLP.** About 1 in 5 eclamptic women have BP below 140/90 at the time of the fit. Judge severity by **symptoms and bloods**, not the BP alone.
- **Oedema is not diagnostic** (most normal pregnant women have ankle oedema), but **sudden facial/hand oedema or rapid weight gain** is a warning.
- **Pre-eclampsia before 20 weeks → think molar pregnancy** (scan, β-hCG) or triploidy.
- **Epigastric pain in a hypertensive pregnant woman is liver involvement until proven otherwise.** Check LFTs and platelets. Antacids delay the diagnosis.
- **Oliguria in pre-eclampsia is usually due to renal vasospasm, not dehydration.** Aggressive fluid challenges drown the patient (pulmonary oedema). Give at most a small bolus (e.g. 250 mL once), then wait; the kidneys usually recover after delivery.
- **The treatment of pre-eclampsia is delivery; the question is always "when and how", balancing maternal risk against fetal prematurity.** At ≥37 weeks the balance always favours delivery.
- **Postpartum does not mean safe.** Many eclamptic fits and strokes occur in the first 48–72 hours after birth. Keep measuring BP, continue $MgSO_4$ for 24 hours and do not stop antihypertensives abruptly.
- **Common student mistakes:** forgetting to check the $MgSO_4$ monitoring chart; recording BP without checking the cuff size; calling 1+ protein "severe"; giving **ergometrine** for the third stage (it causes severe hypertension and stroke); forgetting fetal assessment; forgetting to plan **postnatal BP follow-up and aspirin in the next pregnancy**.

7. Differential Diagnoses

Of hypertension in pregnancy

Differential	Distinguishing clinical feature	Distinguishing investigation
Pre-eclampsia	New BP ≥140/90 after 20 weeks + proteinuria/organ/fetal involvement	Urine PCR/dipstick, FBC, LFT, creatinine, US
Gestational hypertension	New BP after 20 weeks, no proteinuria, normal bloods, normal growth	Same tests are normal; repeat weekly
Chronic hypertension	BP high before 20 weeks or at booking; may have target organ damage	Booking BP records, renal function, ECG
Molar pregnancy	Hypertension/pre-eclampsia before 20 weeks, uterus large for dates, hyperemesis, bleeding	Ultrasound ("snowstorm"), very high β-hCG
Renal disease (glomerulonephritis, lupus nephritis)	Proteinuria/haematuria before 20 weeks, active urinary sediment	Urine microscopy, creatinine, renal US
Phaeochromocytoma (rare)	Paroxysmal hypertension, palpitations, sweating, headaches	Urinary/plasma metanephrines
Thyrotoxicosis	Tachycardia, tremor, weight loss, goitre	TSH, free T4
Pain or anxiety, wrong cuff size	BP normalises with rest and a correct cuff	Repeat BP properly

Of seizures in pregnancy (Uganda)

Differential	Distinguishing clinical feature	Distinguishing investigation
Eclampsia	Hypertension, proteinuria, after 20 weeks or within 6 weeks postpartum	BP, urine protein
Cerebral malaria	Fever, rigors, anaemia, hypoglycaemia; BP usually normal	**mRDT / blood smear**, glucose
Hypoglycaemia	Sweating, known diabetes, malaria on quinine, sepsis	**Bedside glucose**
Epilepsy	Known history, missed antiepileptic drugs, normal BP	History, drug levels
Meningitis / encephalitis	Fever, neck stiffness, photophobia, HIV	LP after stabilisation, CSF CrAg if HIV+
Cerebral venous sinus thrombosis	Postpartum, severe headache, focal signs, papilloedema	CT/MR venography
Stroke (haemorrhagic)	Sudden severe headache, focal deficit, very high systolic BP	CT head
HIV-related CNS disease (toxoplasma, cryptococcus)	HIV+, low CD4, subacute headache	CrAg, CT
Local anaesthetic toxicity, hyponatraemia	After spinal/lignocaine; after large oxytocin infusions in dextrose	Drug history, sodium

Of RUQ pain with abnormal liver tests (the HELLP differential)

Differential	Distinguishing clinical feature	Distinguishing investigation
HELLP	Hypertension (may be mild), haemolysis, low platelets, AST ↑	FBC + film, LDH, LFT
Acute fatty liver of pregnancy	Third trimester, vomiting, jaundice, **hypoglycaemia**, coagulopathy, renal failure; less hypertension	Glucose, PT/INR, bilirubin, ammonia

Differential	Distinguishing clinical feature	Distinguishing investigation
Viral hepatitis (A, B, E)	Very high transaminases (often >1000), jaundice, exposure	Hepatitis serology
Severe malaria with haemolysis	Fever, jaundice, anaemia, thrombocytopenia	mRDT/smear
Cholecystitis / cholangitis	Fever, Murphy's sign, gallstones	Ultrasound
TTP / HUS	Severe thrombocytopenia, haemolysis, neurological signs, renal failure out of proportion; LFTs mild	ADAMTS-13 (rarely available), film

8. Clinical Workup

This is the order in which a senior works up a woman with raised BP after 20 weeks on a Ugandan antenatal or labour ward. The aim is to answer four questions quickly: **Is she fitting or about to fit? Is the BP severe? Which organs are involved? Is the baby coping?** The first 5 minutes are in section 9 (Immediate / emergency).

Bedside

Test	How / what to look for	What it changes	Available at
Urine dipstick	Protein: trace, 1+, 2+, 3+, 4+ (roughly 0.3, 1, 3 and ≥10–20 g/L for 1+ to 4+; varies by brand). **≥2+ is significant; 1+ needs confirming**. Also blood, nitrites/leucocytes (infection or catheter contamination) and ketones	Confirms pre-eclampsia; a **clean-catch midstream** sample avoids false positives from blood, liquor or discharge	All levels (when in stock)
Hb (HemoCue) / packed cell volume	High Hb (haemoconcentration: leaky capillaries) or a falling Hb (haemolysis, abruption)	Severity; blood cross-match	HC III and above
Random blood sugar	<4 mmol/L = hypoglycaemia in pregnancy (some units use <3.0)	Fit differential; **AFLP** (hypoglycaemia + jaundice + coagulopathy)	All levels
mRDT	Positive → treat malaria (IV artesunate if severe) **as well as** $MgSO_4$	Cerebral malaria and eclampsia can coexist	All levels
Bedside clotting test (WHO)	2 mL venous blood in a plain dry glass tube, hold in the fist to keep it warm; tip it gently every minute from 4 minutes. **No clot by 7 minutes, or a soft clot that breaks up = coagulopathy**	Low platelets/DIC → FFP, platelets, avoid spinal anaesthesia; decide urgency	Every level
Hourly urine output (Foley + urometer or measuring jug)	<30 mL/h (<100 mL in 4 h)	Withhold/halve $MgSO_4$; renal involvement; do **not** respond with big fluid boluses	All levels
$MgSO_4$ checks (R-R-U) before every dose	RR ≥16, patellar reflexes present, urine ≥30 mL/h	Give, withhold or reduce the dose	All levels
SpO_2	<95% on air	Pulmonary oedema; stop fluids, furosemide	HC IV and above; often missing at HC III

Test	How / what to look for	What it changes	Available at
FHR (Pinard/Doppler) ± **CTG**	Bradycardia after a fit (usually recovers in 3–5 min), late decelerations, reduced variability	Fetal compromise → expedite delivery once mother is stable	All levels (CTG at referral hospitals, some HC IVs)
Point-of-care ultrasound	Viability, presentation, liquor, placental site, retroplacental clot (a normal scan does **not** exclude abruption)	IUFD → vaginal delivery preferred; breech/transverse → CS planning	HC IV and above with a skilled operator

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Full blood count (Hb, platelets, WBC)	At diagnosis; twice weekly if non-severe; 6–12-hourly in severe disease/HELLP and 24–48 h postpartum	**Platelets <100 × 10^9/L** = severe feature; <150 falling = watch; Hb falling = haemolysis/bleeding	Platelets <100 → HELLP work-up and deliver; **≥75–80** for spinal; **<50 before CS** → transfuse platelets	HC IV (often), all hospitals
Blood film	If platelets low or Hb falling	**Schistocytes** (fragmented red cells), polychromasia	Confirms **microangiopathic haemolysis** (the "H" of HELLP)	Hospital labs
LFTs: AST/ALT, bilirubin	At diagnosis and with any epigastric pain	**AST or ALT >70 IU/L** (≥2× normal) = severe; raised unconjugated bilirubin = haemolysis	Severe features → deliver. Very high (>1,000) → think viral hepatitis	RRH/NRH; some HC IVs
Creatinine, urea, electrolytes	At diagnosis; daily in severe disease; before continuing $MgSO_4$ if oliguric	Normal in pregnancy **35–70 µmol/L**; **>90 µmol/L** = renal involvement; potassium high in AKI	Reduce/withhold $MgSO_4$ (renally excreted); AKI → deliver, strict fluids, nephrology at referral	Hospitals
Uric acid	Optional	Raised	Supports diagnosis only; does **not** decide management	Hospitals
Blood group & cross-match	Everyone with severe disease or planned delivery	2 units ready	Abruption and PPH risk	HC IV with blood bank upward
Blood slide (if mRDT unavailable or negative with fever)	Fever, fit, jaundice	Parasites	Treat malaria in parallel	HC III and above

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Obstetric ultrasound	All: at diagnosis, then every 2 weeks if expectant	EFW <10th centile (FGR), oligohydramnios, presentation, placenta	FGR/oligohydramnios → closer surveillance or delivery; presentation → mode	HC IV (often), hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
Chest X-ray	Breathless, crackles, SpO_2 <95%	Bat-wing shadowing, effusions	Confirms pulmonary oedema (treat clinically first)	Hospitals
CT head	Focal signs, GCS not recovering, fits despite $MgSO_4$, first fit >48 h postpartum	Haemorrhage, infarct, CVT, PRES	Neurosurgical referral; changes the diagnosis	RRH/NRH, private

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
LDH	Suspected HELLP	**≥600 IU/L** = haemolysis	Completes HELLP diagnosis; often unavailable, so use film + bilirubin + falling Hb as proxies	Mainly RRH/NRH and private labs
Clotting: PT/INR, APTT, fibrinogen	Platelets <100, abruption, bleeding, before surgery	Prolonged PT/APTT, **fibrinogen <2 g/L** (normal in pregnancy 4–6)	DIC → FFP, cryoprecipitate; general vs regional anaesthesia	RRH/NRH (bedside clotting test elsewhere)
Urine PCR (or 24-h urine protein)	Dipstick 1+ or equivocal; not needed if dipstick ≥2+	**≥30 mg/mmol** (≥300 mg/24 h)	Confirms proteinuria; amount does **not** grade severity	Referral labs only
Umbilical artery Doppler	FGR or expectant management <34 weeks	Raised resistance; **absent or reversed end-diastolic flow (AEDF/REDF)**	AEDF/REDF → steroids and deliver (usually CS)	RRH/NRH
CTG	In labour or antenatal surveillance	Reduced variability, late decelerations	Fetal compromise → expedite	Referral hospitals, some HC IVs
Serum magnesium	Oliguria, rising creatinine, or signs of toxicity	Therapeutic **2–3.5 mmol/L**	Guides dosing; otherwise rely on clinical R-R-U checks	Rarely available (NRH/private)
sFlt-1/PlGF ratio	Triage only (UK/Europe)	Ratio **≤38** rules out PE for 1 week	Not used in Uganda	Not available in Ugandan public facilities

UGANDA CONTEXT

At an HC III you may have only a **BP machine, a dipstick, a glucometer, an mRDT and a glass tube for the bedside clotting test**. That is enough to diagnose severe pre-eclampsia, give $MgSO_4$ and nifedipine, and refer. **Do not wait for LFTs to refer.** At HC IV you usually add FBC, creatinine, grouping and a scan; LDH, clotting profile and Doppler are mostly at regional and national referral hospitals (Mengo, Mulago, Kawempe).

How often to repeat the tests

Situation	Maternal tests	Fetal tests
Gestational hypertension	BP twice weekly to 4-hourly (inpatient); urine dipstick at each check; FBC, LFT, creatinine at diagnosis and if any symptom	FHR each review; growth scan at diagnosis

Situation	Maternal tests	Fetal tests
Pre-eclampsia without severe features	BP 4-hourly; urine daily; **FBC, LFT, creatinine twice weekly**	Fetal movements chart; FHR each shift; **growth and liquor scan every 2 weeks**; Doppler where available
Severe pre-eclampsia / on $MgSO_4$	BP, pulse every 15–30 min until stable, then hourly; **RR, reflexes, urine output hourly** (before every IM dose at least); bloods **daily**, or **6–12-hourly** if HELLP	FHR every 30 min (continuous CTG where available)
Postpartum	BP 4-hourly for at least 3 days; repeat FBC, LFT, creatinine **24–48 h after delivery**	Neonatal review ($MgSO_4$ hypotonia, prematurity)

Putting it together

Model one-line summary

> "A 24-year-old G2P1 at 33 weeks with **pre-eclampsia with severe features** (BP 168/112, headache, platelets 88), currently on Pritchard $MgSO_4$ with R-R-U checks normal, BP controlled to 148/96 on nifedipine, fetus alive and cephalic with EFW on the 8th centile, for steroids and delivery."

Problem list

1. Severe hypertension: controlled / uncontrolled (what drugs, how many doses).
2. Seizure prophylaxis: $MgSO_4$ started at (time), next dose due at (time), toxicity checks.
3. Organ involvement: blood (platelets), liver (AST/ALT, pain), kidney (creatinine, urine output), brain (symptoms, clonus), lungs (SpO_2).
4. Fetus: GA, EFW/FGR, liquor, Doppler/FHR, steroids given?
5. Delivery plan: timing, mode, Bishop score, anaesthesia (platelets), blood available.
6. Postpartum plan: $MgSO_4$ for 24 h, BP, repeat bloods, oxytocin (not ergometrine).

Working diagnosis: name the category precisely: "pre-eclampsia **with/without** severe features" ± "HELLP" ± "eclampsia", with the gestation and fetal status. If an alternative is still open (malaria, AFLP), say so.

Model case presentation (ward round)

> "Mrs N.A., a 19-year-old primigravida at 35 weeks by a 12-week scan, admitted 6 hours ago from Kasangati HC IV with **pre-eclampsia with severe features**: BP 170/115, severe frontal headache and blurred vision, and 3+ proteinuria. She received a Pritchard loading dose of $MgSO_4$ at 10:00 and hydralazine 5 mg IV twice. Currently BP 150/100, RR 18, reflexes present, urine output 40 mL/h, SpO_2 97%. No clonus. Platelets 140, AST 45, creatinine 70. The fetus is cephalic, FHR 142, EFW 2.1 kg (10th centile). Bishop score 4. **Plan**: continue $MgSO_4$ with 4-hourly monitoring, nifedipine MR 20 mg 12-hourly, cervical ripening with misoprostol 25 mcg for delivery within 24 h, oxytocin (not ergometrine) for the third stage, and continue $MgSO_4$ for 24 h postpartum."

Worked example 1: a PET bloods panel

Result	Value	Interpretation
Hb	9.6 g/dL (was 11.8 last week)	Falling: haemolysis or bleeding
Platelets	64×10^9/L	**<100 = severe**; below 75–80 → spinal not advised
AST / ALT	186 / 140 IU/L	**>70 = elevated liver enzymes**
LDH	910 IU/L	**≥600 = haemolysis**
Bilirubin	32 µmol/L (mainly unconjugated)	Haemolysis

Result	Value	Interpretation
Creatinine	98 µmol/L	**>90 = renal involvement**
Glucose	5.2 mmol/L	Normal: argues against AFLP
Film	Schistocytes	Microangiopathy

Reading it: all three HELLP criteria (H: LDH ≥600 + schistocytes; EL: AST ≥70; LP: platelets <100) are met = **complete HELLP**, with renal involvement. Normal glucose and only mildly raised bilirubin make AFLP less likely; transaminases in the hundreds (not thousands) make viral hepatitis less likely. **Action:** treat as severe PE ($MgSO_4$, BP control; check urine output before each dose because of the raised creatinine), **deliver** (after steroids only if <34 weeks and she is stable, within 24–48 h), group and cross-match, platelets ready, bedside clotting test/clotting profile, **general anaesthesia or platelet transfusion** if CS is needed, repeat bloods 6–12-hourly.

Worked example 2: $MgSO_4$ toxicity check at the 4-hourly dose

Check	Found	Rule
RR	12/min	**<16 → withhold**
Patellar reflexes	Absent	**Absent → withhold**
Urine output	60 mL in 4 h	**<100 mL in 4 h → withhold/reduce**

Reading it: this is **magnesium toxicity** until proven otherwise, probably from reduced renal excretion (oliguria). **Action:** do **not** give the dose; call for help; oxygen, check airway and SpO_2; **calcium gluconate 1 g (10 mL of 10%) IV over 10 min** if breathing is depressed or stops (WHO MCPC specifies it for respiratory arrest; many units give it once RR is falling below 12, so follow your protocol), with bag-mask ventilation if needed; check creatinine and serum Mg if available. Restart $MgSO_4$ (often at a reduced dose) only when reflexes return, RR ≥16 and urine output ≥30 mL/h. If she has had a spinal, test the **biceps** reflex instead.

Worked example 3: reading the urine

- Dipstick **1+** on a midstream sample, BP 146/94 at 32 weeks, bloods normal → **gestational hypertension with possible proteinuria**: confirm with a **urine PCR** (≥30 mg/mmol = pre-eclampsia) or repeat dipstick. If PCR is unavailable, many Ugandan units repeat a clean-catch dipstick and treat persistent 1+ with caution.
- Dipstick **3+** with BP 150/100 and no symptoms, normal bloods → **pre-eclampsia without severe features** (3+ does **not** make it severe on modern criteria), for monitoring and delivery at 37 weeks.
- Dipstick **1+** with platelets 70 → **pre-eclampsia with severe features** (the platelets decide severity, not the protein).

9. Management

Management rests on **five pillars**: (1) control severe hypertension, (2) prevent/treat seizures with $MgSO_4$, (3) careful fluid balance, (4) assess mother and fetus, and (5) decide the timing and mode of delivery. Then (6) care for her postpartum and plan for the future.

Immediate / emergency

First 5 minutes (any woman with raised BP after 20 weeks)

1. **Is she fitting, post-ictal or drowsy?** If yes, this is **eclampsia until proven otherwise**: follow the eclampsia algorithm below.
2. **Vital signs**: BP (correct cuff, seated or left lateral), pulse, **RR**, **SpO_2**, temperature, GCS. Count the **RR before any $MgSO_4$** so you have a baseline.

3. **Is the BP severe (≥160 systolic and/or ≥110 diastolic)?** Repeat within **15 minutes**. If confirmed, give an **acute antihypertensive now** (nifedipine IR 10 mg orally or hydralazine 5 mg IV slowly) with the aim of treating within **30–60 minutes**.
4. **Ask the three "imminent eclampsia" questions**: headache? visual disturbance? epigastric/RUQ pain? And **check reflexes and clonus**. Any one of these with hypertension = severe features → **$MgSO_4$ now**.
5. **Quick look for the killers**: breathlessness/SpO_2 <95%/crackles (**pulmonary oedema**), vaginal bleeding with a hard tender uterus (**abruption**), bruising or oozing from puncture sites (**DIC**).
6. **Two bedside checks for every fit**: **blood glucose** and **mRDT** (hypoglycaemia and cerebral malaria are the great Ugandan mimics).
7. **Start the bundle**: IV access (16–18 G), draw bloods, **catheter with hourly urine output**, fluid restriction to **80 mL/h**, calcium gluconate at the bedside, and **FHR**.
8. **Call**: the senior midwife and medical officer immediately; the obstetrician and anaesthetist if severe or eclamptic. At HC II/III: loading dose + first antihypertensive, then **refer with an escort and the next 5 g IM dose**.

Eclampsia: emergency algorithm (ABCDE)

1. **Call for help. Do not leave her alone.** Note the time.
2. **Protect from injury** (sides of bed up, remove sharp objects). **Do not restrain** her. **Do not put anything in her mouth.**
3. **A**: after the fit, **left lateral (recovery) position**, clear the airway, **suction**.
4. **B**: **oxygen 4–6 L/min** by mask; check breathing; bag-mask ventilation if not breathing.
5. **C**: check pulse; **IV access** (16–18 G); take bloods (FBC, LFT, creatinine, clotting, glucose, mRDT, group & cross-match).
6. **$MgSO_4$ loading dose** immediately (Pritchard 4 g IV + 10 g IM), then maintenance.
7. **D**: **check blood glucose** (hypoglycaemia) and **mRDT** (malaria). Treat what you find. GCS.
8. **Treat BP** if ≥160/110 (nifedipine IR 10 mg oral once she can swallow safely, otherwise hydralazine 5 mg IV).
9. **Catheterise**, strict input/output, **restrict fluids** to 80 mL/h.
10. **E / fetus**: **assess the fetus.** Expect fetal bradycardia during and just after a fit. It usually recovers within 3–5 minutes once the mother is oxygenated. **Do not rush to theatre during the bradycardia; resuscitate the mother first.**
11. **Deliver within 12 hours** of the fit once she is stable, regardless of gestation.
12. Monitor: BP, pulse, RR, SpO_2, reflexes, urine output, GCS every 15–30 min; and document everything on an eclampsia chart.
13. Postpartum: HDU/ICU if unconscious; continue $MgSO_4$ for 24 h after delivery or last fit.

Definitive

Gestational hypertension and pre-eclampsia WITHOUT severe features

1. **Admit** at least for initial assessment (in Uganda most units admit all women with pre-eclampsia, because follow-up and transport are unreliable).
2. BP 4-hourly, urine protein daily, fetal movements chart, FHR every shift.
3. Bloods (FBC, LFT, creatinine) at diagnosis and then **twice weekly**; ultrasound for growth and liquor every 2 weeks.
4. Start an **oral antihypertensive if BP is persistently ≥140/90** (CHAP trial and NICE: target about **135/85**; do not drop diastolic below 80).
5. No $MgSO_4$ routinely (unless severe features appear or during labour/after delivery per local protocol).
6. **Plan delivery at 37+0 weeks** (HYPITAT trial: induction at ≥37 weeks reduces maternal complications without increasing CS). Deliver earlier if severe features, fetal compromise or maternal request after counselling.

Pre-eclampsia WITH severe features: step by step

1. **Call for help**: senior midwife, medical officer or obstetrician, and the anaesthetist if delivery is likely.
2. **Admit to a high-dependency bed** close to the nursing station, in a quiet room. Left lateral position.
3. **IV access** with a large-bore cannula (16–18 G). Draw FBC, LFT, LDH, creatinine, clotting, group & cross-match, glucose and mRDT.
4. **Control BP within 30–60 minutes** if ≥160/110 (drug table). Aim for **140–150 / 90–100**. Avoid a precipitous fall, which reduces placental perfusion and causes fetal distress.
5. **Give $MgSO_4$** (Pritchard loading dose, or Zuspan where infusion pumps are available).
6. **Insert a Foley catheter** and start a strict **input/output chart**; urine output hourly.
7. **Restrict fluids**: total IV plus oral intake about **80 mL/h (1 mL/kg/h)**. No routine fluid boluses.
8. **Assess the fetus**: FHR, ultrasound for growth, liquor and presentation; Doppler if available.
9. If **24 (Uganda: 28) to 34 weeks**, give **antenatal corticosteroids** (dexamethasone 6 mg IM 12-hourly × 4), but **do not delay delivery for steroids** when the mother is unstable.
10. **Plan delivery** (timing below). Inform the neonatal team.
11. Monitoring every 15–30 minutes until stable, then hourly: BP, pulse, RR, SpO_2, reflexes, urine output, symptoms, FHR.

Magnesium sulphate: indications, preparations, regimens, monitoring

Indications: eclampsia (treatment); pre-eclampsia with severe features (prevention); and many units also give it intrapartum and for 24 h postpartum to all women with pre-eclampsia. It is also used for **fetal neuroprotection before 32 weeks** (see the preterm labour topic).

Preparations (know these by heart)

- $MgSO_4$ comes as **50% solution** (5 g in 10 mL, i.e. 0.5 g per mL). Check the ampoule strength: 10 mL of 50% = 5 g.
- **20% solution for IV use** = 4 g (8 mL of 50%) + 12 mL of water for injection or normal saline = **20 mL of 20%**.

Regimen	Loading dose	Maintenance	Duration
Pritchard (IM), the standard in Uganda	**4 g of 20% IV slowly over 5–10 min** (20 mL) **PLUS 10 g of 50% IM**: 5 g (10 mL) deep IM into **each upper outer buttock**, each with **1 mL of 2% lignocaine** in the same syringe. Total loading = **14 g**	**5 g of 50% IM** (+ 1 mL 2% lignocaine) **every 4 hours in alternate buttocks**	**24 hours after delivery or after the last fit, whichever is later**
Zuspan (IV), where infusion pumps and staff allow	**4 g IV over 5–20 min** (some protocols 20–30 min)	**1 g/hour by continuous IV infusion** (e.g. 5 g in 250 mL normal saline at 50 mL/h via pump)	Same: 24 h after delivery or last fit
Recurrent fit on $MgSO_4$	**2 g of 20% IV over 5 min** (4 g if weight >70 kg per some protocols), in addition to the regimen	Continue	Look for another cause if fits persist (CT)
At HC II/III before referral	Give the **loading dose** (IM 10 g ± IV 4 g if trained)	Refer immediately with an escort, the treatment sheet and the next dose (5 g IM)	Next dose due 4 h after loading, even on the road

DRUG DOSING

Checks before EVERY maintenance dose (and hourly on an infusion):

1. **Respiratory rate ≥16/min**
2. **Patellar reflexes present** (use biceps reflex if she has had spinal/epidural anaesthesia)
3. **Urine output ≥30 mL/h** (≥100 mL in the previous 4 hours)

- **RR <16 or reflexes absent → withhold** the dose and call for help.
- **Urine <30 mL/h** → withhold further IM doses (or halve the infusion to 0.5 g/h), check creatinine, and resume when urine output improves.
- **Respiratory arrest** → ventilate with bag and mask, give **calcium gluconate 1 g (10 mL of 10%) IV slowly over 10 minutes**, stop $MgSO_4$.
- Keep **calcium gluconate drawn up/at the bedside** throughout.

Side effects to warn about: warmth and flushing during the IV loading (normal), nausea, and pain at the IM site (hence lignocaine). The neonate may be floppy (hypotonia), so tell the resuscitation team.

If $MgSO_4$ is not available: **diazepam** (inferior; use only if no $MgSO_4$): 10 mg IV slowly over 2 minutes, repeat once if fits recur; maintenance 40 mg in 500 mL normal saline titrated to keep her sedated but rousable (max 100 mg in 24 h). Risk of respiratory depression in mother and neonate. Rectal diazepam 20 mg if no IV access.

HELLP syndrome

1. Treat as **severe pre-eclampsia**: $MgSO_4$, BP control, fluid restriction.
2. **Deliver**: immediately if ≥34 weeks, DIC, liver haematoma, abruption or fetal compromise. Between 24/28 and 34 weeks, deliver after steroids (within 24–48 h) if the mother is stable.
3. **Platelets**:
   - Transfuse platelets if <20 × 10^9/L, or **<50 × 10^9/L before caesarean** or if actively bleeding.
   - Spinal anaesthesia is generally safe with platelets **≥75–80 × 10^9/L** and a stable count. Below that, general anaesthesia (with careful blunting of the intubation pressor response).
4. Correct coagulopathy with **fresh frozen plasma**; fibrinogen with cryoprecipitate if available.
5. Watch for **liver haematoma/rupture** (shoulder-tip pain, shock): ultrasound, surgical team.
6. Dexamethasone to improve the platelet count is **not** recommended routinely (no outcome benefit), only for fetal lung maturity.
7. Labs 6–12-hourly until improving; the nadir is usually 24–48 h postpartum.

Timing and mode of delivery

Situation	Timing
Eclampsia	Deliver **within 12 hours** of the fit, after stabilisation
Severe pre-eclampsia **≥34 weeks**	Deliver once stable (within 24 h)
Severe pre-eclampsia **<34 weeks** (viable)	Stabilise, give steroids, and deliver after 48 h if stable, **or immediately** if: uncontrollable BP despite 3 drugs, eclampsia, pulmonary oedema, HELLP/DIC, rising creatinine, abruption, non-reassuring fetal status, IUFD. Expectant management only in a referral centre with NICU and daily review
Severe pre-eclampsia **before viability** (Uganda <28 weeks)	Counsel; maternal life comes first. Termination of pregnancy is usually advised

Situation	Timing
Pre-eclampsia without severe features **<37 weeks**	Expectant management with monitoring; deliver at 37+0 or earlier if it becomes severe
Pre-eclampsia without severe features **≥37 weeks**	**Deliver** (induction)
Gestational hypertension, well controlled	Deliver at 37–39 weeks
Chronic hypertension, well controlled	Deliver at 37–39 weeks

Mode of delivery

- **Vaginal delivery is preferred** if there are no obstetric contraindications: less blood loss and no anaesthetic risk.
- Cervix favourable (Bishop ≥6–8): **ARM + oxytocin**.
- Cervix unfavourable: **misoprostol 25 mcg vaginally every 6 h** or **25 mcg orally every 2 h**, or a Foley balloon catheter.
- **Caesarean section** if: usual obstetric indications; fetal distress; the cervix is very unfavourable and delivery is not expected within 12 h (eclampsia) or 24 h (severe pre-eclampsia); severe disease far from term with a compromised fetus.
- **Second stage**: consider **shortening it with vacuum** if BP is uncontrolled or she is exhausted (bearing down raises BP and intracranial pressure).
- **Third stage**: **oxytocin 10 IU IM** is the uterotonic. **NEVER ergometrine or Syntometrine**: it causes vasoconstriction, a hypertensive crisis and stroke.
- **Anaesthesia**: spinal is preferred (platelets ≥75–80). General anaesthesia risks a hypertensive surge at intubation (stroke) and a difficult airway (laryngeal oedema).

Drug table

Acute antihypertensives for severe hypertension (≥160/110)

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
Nifedipine immediate-release	**10 mg**	**Oral (swallow; do not bite or give sublingually)**	Recheck BP after 20–30 min; repeat 10–20 mg if BP still ≥160/110	Up to about 30–40 mg in the first hour; then continue MR 20 mg 12-hourly	First choice where there is no IV access or pump; most available in Uganda. Headache and tachycardia
Hydralazine	**5 mg** (10 mg if very high)	**IV slowly over 2–5 min**	Check BP every 5 min for 20 min; repeat 5–10 mg every 20–30 min	Max **20 mg IV** per episode (WHO alternative: 12.5 mg IM every 2 h)	Dilute the 20 mg ampoule to 20 mL with water for injection or normal saline → 1 mg per 1 mL (so 5 mg = 5 mL). Tachycardia, headache (mimics imminent eclampsia), sudden hypotension → fetal distress
Labetalol	**20 mg**	**IV over 2 min**	Then 40 mg, 80 mg, 80 mg every 10 min if needed	Max **300 mg**; or infusion 20–160 mg/h	Avoid in asthma and heart failure. Rarely available in Ugandan public facilities

Oral (maintenance) antihypertensives

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
Methyldopa	250 mg, increase to 500–750 mg	Oral	8-hourly	Max 3 g/day; switch to another agent within 2 days of delivery	Centrally acting; safest long-term data. **Slow onset (hours to days, 4–6 h): never for acute control**. Sedation, **depression**, postural hypotension, positive Coombs, hepatitis
Nifedipine modified-release (retard)	20 mg, increase to 40 mg	Oral	12-hourly	Max 120 mg/day	Calcium-channel blocker. Headache, flushing, ankle oedema. Widely available in Uganda
Labetalol (oral)	100–200 mg, up to 400 mg	Oral	8–12-hourly	Max 2.4 g/day	α + β blocker. First line in the UK. **Avoid in asthma**. Scarce in Uganda
Amlodipine	5–10 mg	Oral	Once daily		Alternative calcium-channel blocker; postnatal first choice
Enalapril (postnatal only)	5–20 mg	Oral	Once daily		Safe in breastfeeding; check creatinine and potassium. **Contraindicated in pregnancy**

Anticonvulsants, antidote and other drugs

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
$MgSO_4$ Pritchard	Load **4 g of 20% IV** over 5–10 min + **10 g of 50% IM** (5 g each buttock with 1 mL 2% lignocaine)	IV + deep IM	Maintenance **5 g of 50% IM every 4 h**, alternate buttocks	24 h after delivery or last fit	R-R-U checks before every dose
$MgSO_4$ Zuspan	Load **4 g IV** over 5–20 min	IV	**1 g/h** infusion (0.5 g/h if oliguric)	24 h after delivery or last fit	Hourly R-R-U checks; pump needed
$MgSO_4$ for recurrent fit	**2 g of 20% IV** over 5 min (4 g if >70 kg per some protocols)	IV	Once		Look for another cause if fits persist
Calcium gluconate (antidote)	**1 g (10 mL of 10%)**	IV slowly	Over 10 min	Once, repeat per protocol	Keep at the bedside
Diazepam (only if no $MgSO_4$)	**10 mg**; rectal 20 mg if no IV	IV over 2 min	Repeat once if fits recur; maintenance 40 mg in 500 mL NS titrated	Max 100 mg/24 h	Respiratory depression in mother and neonate
Dexamethasone	**6 mg**	IM	12-hourly	× 4 doses	24 (Uganda 28) to 34 weeks; do not delay delivery for steroids if unstable
Furosemide	**20–40 mg**	IV	Once, repeat per response		Pulmonary oedema only; diuretics otherwise reduce plasma volume
Misoprostol (induction)	**25 mcg**	Vaginal / oral	Every 6 h vaginal or every 2 h oral	Per induction protocol	Unfavourable cervix

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
Oxytocin (third stage)	**10 IU**	IM	Once		Never ergometrine or Syntometrine
Aspirin (prevention)	**75–150 mg**	Oral	Once daily at night	12 → 36 weeks	High-risk women
Calcium (prevention)	**1.5–2.0 g** elemental	Oral	Daily in 3 divided doses	20 weeks to delivery	Take apart from iron
Enoxaparin	Per weight Verify *dose not stated in source*	SC	Daily	Postpartum, per risk	Only once platelets are safe

DRUG DOSING

Methyldopa has **no role** in acute severe hypertension (onset 4–6 hours). Give nifedipine IR or hydralazine, then add methyldopa or nifedipine MR for maintenance. Nifedipine and $MgSO_4$ can be given together; the feared interaction (profound hypotension/neuromuscular block) is rare in practice. Just monitor.

DRUG DOSING

Contraindicated in pregnancy: ACE inhibitors (enalapril, captopril) and ARBs (losartan) cause fetal renal failure, oligohydramnios and skull defects; stop them and switch before or as soon as pregnancy is confirmed. **Atenolol** is linked to FGR. **Diuretics** reduce plasma volume and are used only for pulmonary oedema. ACE inhibitors become **first choice again after delivery** (enalapril is compatible with breastfeeding).

UGANDA CONTEXT

The common Ugandan combination on the labour ward is **hydralazine IV (or nifedipine oral) for acute control + $MgSO_4$ Pritchard + methyldopa ± nifedipine retard for maintenance**. Where hydralazine is out of stock, oral nifedipine is as effective (RCT evidence shows nifedipine, labetalol and hydralazine are all acceptable). Follow your unit protocol.

Referral criteria

Level	What to do	Refer up when
HC II/III	Detect (BP + urine at every ANC visit). For severe features or eclampsia: $MgSO_4$ **loading dose**, first dose of antihypertensive (nifedipine IR 10 mg), catheterise if possible	**Immediately** for severe features or eclampsia → HC IV/hospital, with a health worker, notes and the next $MgSO_4$ dose
HC IV / General Hospital	Full $MgSO_4$ regimen, IV hydralazine or oral nifedipine, steroids, basic labs (FBC, dipstick, clotting time), induction or **caesarean section**, HDU-type monitoring	HELLP with bleeding, unconscious after eclampsia, pulmonary oedema needing ventilation, or <34 weeks needing NICU → RRH/NRH
RRH / NRH (Mengo, Mulago, Kawempe)	ICU/HDU, blood products (platelets, FFP), CT, NICU for preterm babies, expectant management of early-onset disease, anaesthetic expertise	n/a

Monitoring and follow-up

- **On $MgSO_4$**: BP, pulse every 15–30 min until stable, then hourly; RR, reflexes, urine output hourly (before every IM dose at least); SpO_2; symptoms; FHR every 30 min.
- **Bloods**: daily in severe disease; 6–12-hourly in HELLP; repeat **FBC, LFT and creatinine 24–48 h after delivery** (HELLP can appear or worsen postpartum).
- **Discharge** when BP <150/100 on oral drugs, symptoms have resolved and bloods are improving. **Review BP within 1 week**, then at 6 weeks.
- Refer if hypertension/proteinuria persists beyond 6–12 weeks (chronic hypertension or renal disease).

Postpartum care

1. **Continue $MgSO_4$ for 24 hours** after delivery (or after the last fit).
2. **BP 4-hourly** (more often if severe) for at least 3 days. BP typically peaks on **days 3–6**.
3. Continue antihypertensives and **wean slowly** as BP falls. After delivery:
   - Switch **methyldopa** to another agent within 2 days (risk of **postnatal depression**).
   - First choices postnatally: **nifedipine MR 20 mg 12-hourly**, **amlodipine 5–10 mg daily**, or **enalapril 5–20 mg daily** (safe in breastfeeding; check creatinine and potassium).
   - Target <140/90.
4. Continue **fluid balance charting** until urine output normalises (≥0.5 mL/kg/h). Watch for **pulmonary oedema** (mobilisation of oedema fluid). If it develops: sit up, oxygen, **furosemide 20–40 mg IV**, stop IV fluids.
5. **Analgesia**: paracetamol is fine. NSAIDs are best avoided if there is renal impairment, oliguria or low platelets (otherwise ACOG considers them acceptable).
6. **Thromboprophylaxis** (enoxaparin) if CS, prolonged immobility or other risk factors, once platelets are safe.
7. **Future counselling**:
   - **Recurrence** about 15% (up to 25–65% after early severe disease or HELLP).
   - **Aspirin 75–150 mg from 12 weeks** in every future pregnancy.
   - **Lifelong cardiovascular risk** (double the risk of hypertension, heart disease and stroke): annual BP checks.
8. **Contraception**: progestogen-only methods, IUD and implants are safe. **Avoid combined oestrogen methods** while hypertensive (UKMEC/WHO MEC category 3–4).

10. Complications

Disease-related

Maternal	Fetal / neonatal
Eclampsia; **intracerebral haemorrhage** (the leading cause of death); PRES	**Fetal growth restriction**, oligohydramnios
Pulmonary oedema, ARDS	**Placental abruption** → hypoxia, IUFD
HELLP, DIC, liver haematoma/rupture	**Iatrogenic prematurity** (RDS, IVH, NEC, sepsis)
Acute kidney injury	Stillbirth
Placental abruption, PPH	Birth asphyxia
Cortical blindness, retinal detachment (usually reversible)	Long-term: cardiometabolic risk of FGR
Aspiration pneumonia (after fits)	
Long-term: chronic hypertension, ischaemic heart disease, stroke, CKD	

Treatment-related

Complication	Cause	Action
Loss of reflexes, respiratory depression, cardiac arrest	$MgSO_4$ accumulation (oliguria)	Withhold; calcium gluconate; ventilate
Neonatal hypotonia/respiratory depression	$MgSO_4$; diazepam	Warn the neonatal team
Precipitous BP fall, fetal distress	Over-rapid antihypertensive (hydralazine)	Titrate; target 140–150/90–100
Pulmonary oedema	IV fluid boluses for oliguria	Restrict to 80 mL/h
Hypertensive crisis, stroke	Ergometrine/Syntometrine	Use oxytocin only
Postnatal depression	Methyldopa	Switch within 2 days of delivery
Fetal renal failure, oligohydramnios, skull defects	ACE inhibitors/ARBs in pregnancy	Stop and switch
FGR	Atenolol	Avoid
Spinal/epidural haematoma	Regional anaesthesia with platelets <75–80	GA or platelet transfusion
Hypertensive surge at intubation, difficult airway	General anaesthesia	Blunt the pressor response; senior anaesthetist

11. Prognosis

- **Delivery cures it**, but the disease can worsen for 48–72 hours postpartum; BP peaks on days 3–6.
- Cortical blindness and retinal detachment are usually reversible; AKI usually recovers after delivery.
- **Recurrence** about 15% (up to 25–65% after early severe disease or HELLP).
- **Lifelong cardiovascular risk**: double the risk of hypertension, heart disease and stroke; CKD.
- Babies: risks of prematurity and FGR (including long-term cardiometabolic risk).

12. Prevention

Intervention	Who	Dose and timing
Low-dose aspirin	One high-risk factor, or two or more moderate risk factors	**75–150 mg orally once daily at night**, start at **12 weeks** (ideally before 16, and before 20 weeks at the latest), continue to **36 weeks**
Calcium supplementation	All pregnant women in populations with low dietary calcium (most of Uganda)	**1.5–2.0 g elemental calcium daily** in 3 divided doses from 20 weeks to delivery (WHO). Take it apart from iron tablets (calcium blocks iron absorption)
BP and urine at every ANC visit	Everyone	Early detection
Health education on danger signs	Everyone	Headache, blurred vision, epigastric pain, fits, swelling of the face → come immediately

UGANDA CONTEXT

In Uganda, aspirin is available mainly as 75 mg tablets (many specialists give 2 × 75 mg = 150 mg at night for high-risk women, following FIGO/ASPRE). Calcium is not yet reliably supplied through the public ANC package, so counsel on dietary calcium (milk, *mukene*/silverfish, dark green leaves) and prescribe it where the family can buy it.

Secondary and tertiary prevention: $MgSO_4$ prevents eclampsia in severe disease (Magpie: halves the risk); prompt treatment of severe BP prevents stroke; postnatal BP follow-up and aspirin in every future pregnancy.

13. Special Populations

- **Asthma or heart failure**: avoid labetalol; use nifedipine or hydralazine. Caution with fluids in heart disease.
- **Chronic hypertension**: stop ACE inhibitors/ARBs/atenolol before or as soon as pregnancy is confirmed; watch for superimposed pre-eclampsia (about 25%).
- **Before 20 weeks**: think molar pregnancy or triploidy.
- **Preterm (<34 weeks)**: steroids; $MgSO_4$ also gives fetal neuroprotection <32 weeks; expectant management only at a referral centre with NICU.
- **After spinal anaesthesia**: test the biceps reflex for $MgSO_4$ monitoring.
- **Renal impairment / oliguria**: reduce or withhold $MgSO_4$; avoid NSAIDs.
- **Co-existing malaria**: treat both (IV artesunate if severe, and $MgSO_4$).
- **HIV**: widen the seizure differential (toxoplasma, cryptococcus; CSF CrAg).
- **Breastfeeding**: enalapril, nifedipine and amlodipine are compatible.

14. Key Points

High-yield points

- New BP ≥140/90 after 20 weeks + proteinuria **or** organ dysfunction **or** FGR = pre-eclampsia; proteinuria is not required.
- Severity is judged by BP ≥160/110, symptoms, platelets, liver, kidney, lungs and fetus, not by the pluses of protein.
- BP ≥160/110 confirmed within 15 min: treat within 30–60 min (nifedipine IR 10 mg oral or hydralazine 5 mg IV); target 140–150/90–100.
- $MgSO_4$ Pritchard 4 g IV + 10 g IM, then 5 g IM 4-hourly for 24 h after delivery or last fit.
- R-R-U before every dose; calcium gluconate 1 g IV over 10 min is the antidote.
- A fit after 20 weeks is eclampsia until proven otherwise; check glucose and mRDT.
- Fluids 80 mL/h; oliguria is vasospasm, not dehydration.
- Deliver: eclampsia within 12 h, severe ≥34 weeks within 24 h, non-severe at 37 weeks.
- Oxytocin, never ergometrine, for the third stage.
- Postpartum is not safe: BP peaks days 3–6; review within 1 week; aspirin next pregnancy.

Numbers & doses

Item	Value
Hypertension in pregnancy	**≥140/90** on two readings (4 h apart if non-severe)
Severe hypertension	**≥160/110**, confirmed within 15 min; treat within **30–60 min**

Item	Value
Acute BP target	**140–150 / 90–100**; maintenance ~**135/85**; diastolic not below 80
Significant proteinuria	Dipstick **≥2+**; PCR **≥30 mg/mmol**; **≥300 mg/24 h**
HELLP thresholds	LDH **≥600 IU/L**, AST **≥70 IU/L**, platelets **<100 × 10⁹/L**
Renal severe feature	Creatinine **>90 µmol/L** (pregnancy normal 35–70); urine **<30 mL/h**
Nifedipine IR (acute)	**10 mg orally**, repeat 10–20 mg after 20–30 min; ~30–40 mg in the first hour
Hydralazine (acute)	**5 mg IV slowly**, repeat 5–10 mg every 20–30 min; max **20 mg** per episode (check local protocol)
Labetalol IV	**20 mg**, then 40, 80, 80 mg every 10 min; max **300 mg**
Methyldopa	250–750 mg **8-hourly**, max 3 g/day (maintenance only)
Nifedipine MR	20–40 mg **12-hourly**, max 120 mg/day
$MgSO_4$ Pritchard loading	**4 g of 20% IV** over 5–10 min + **10 g of 50% IM** (5 g each buttock + 1 mL 2% lignocaine) = 14 g
$MgSO_4$ Pritchard maintenance	**5 g of 50% IM every 4 h**, alternate buttocks
$MgSO_4$ Zuspan	**4 g IV** over 5–20 min, then **1 g/h**
Recurrent fit on $MgSO_4$	**2 g of 20% IV** over 5 min
$MgSO_4$ duration	**24 h after delivery or last fit**, whichever is later
Making 20% $MgSO_4$	**8 mL of 50% + 12 mL water** = 20 mL (4 g)
$MgSO_4$ checks	RR **≥16**, reflexes present, urine **≥30 mL/h** (≥100 mL/4 h)
Serum Mg	Therapeutic **2–3.5 mmol/L**; reflexes lost 4–5; respiratory depression 6–7.5; arrest >12
Antidote	**Calcium gluconate 1 g (10 mL of 10%) IV over 10 min**
Diazepam (only if no $MgSO_4$)	**10 mg IV** over 2 min; rectal 20 mg
Fluid limit	**80 mL/h (1 mL/kg/h)** total
Steroids (24/28–34 weeks)	**Dexamethasone 6 mg IM 12-hourly × 4**
Platelets and anaesthesia	Spinal **≥75–80**; transfuse **<50 before CS**, **<20** always
Delivery timing	Eclampsia **≤12 h**; severe PE ≥34 weeks **≤24 h**; non-severe **37+0 weeks**
Bedside clotting test	No clot by **7 min** = coagulopathy
Aspirin prevention	**75–150 mg nocte**, 12 → 36 weeks
Calcium prevention	**1.5–2 g/day** elemental from 20 weeks
Postpartum BP peak	Days **3–6**; review within **1 week** of discharge
Recurrence	About **15%** (higher after early severe disease/HELLP)

Exam traps

- **"3+ proteinuria = severe pre-eclampsia."** Wrong: severity is judged by BP ≥160/110, symptoms, bloods, lungs and fetus, not the number of pluses.
- **"No proteinuria, so it cannot be pre-eclampsia."** Wrong: new hypertension + low platelets, raised transaminases/creatinine, neurological signs or FGR is pre-eclampsia.
- **"Give methyldopa for BP 170/115."** Wrong: methyldopa takes hours. Give **nifedipine IR 10 mg orally or hydralazine 5 mg IV**.

- **"$MgSO_4$ is the antihypertensive."** Wrong: it prevents fits; you still need an antihypertensive to prevent stroke.
- **"Diazepam first for an eclamptic fit."** Wrong: **$MgSO_4$** is first line; diazepam only if $MgSO_4$ is unavailable.
- **"She is oliguric, give 2 L of fluid."** Wrong: oliguria is renal vasospasm. Restrict to 80 mL/h (at most one small bolus); pulmonary oedema kills.
- **"Rush to caesarean for fetal bradycardia during the fit."** Wrong: stabilise the mother first; FHR usually recovers in 3–5 min.
- **"Ergometrine for the third stage."** Wrong: **oxytocin 10 IU IM**; ergometrine causes hypertensive crisis and stroke.
- **"BP is only 140/90, so it can't be eclampsia."** About 1 in 5 eclamptic women have BP below 140/90 at the fit.
- **"Epigastric pain = gastritis, give an antacid."** It is liver involvement until proven otherwise.

Questions seniors ask

Viva questions

1. **What are the monitoring parameters for $MgSO_4$?** RR ≥16, patellar reflexes present, urine ≥30 mL/h; serum Mg if available (2–3.5 mmol/L). Antidote: calcium gluconate 1 g IV over 10 min.
2. **Why lignocaine with the IM $MgSO_4$?** The 50% solution is very painful. 1 mL of 2% lignocaine in the same syringe reduces the pain.
3. **Why no ergometrine?** It causes smooth-muscle vasoconstriction → a sudden rise in BP → stroke or eclampsia.
4. **Why restrict fluids when she is oliguric?** Capillary leak and low oncotic pressure mean fluids go into the lungs, not the circulation. Oliguria reflects renal vasospasm; it resolves after delivery.
5. **When is proteinuria not needed to diagnose pre-eclampsia?** When there is new hypertension plus organ dysfunction (low platelets, raised creatinine/transaminases, neurological symptoms, pulmonary oedema) or FGR.
6. **What is the pathophysiological basis?** Failed trophoblast invasion/spiral artery remodelling → placental ischaemia → sFlt-1/soluble endoglin → systemic endothelial dysfunction.
7. **How long is $MgSO_4$ continued?** 24 hours after delivery or the last fit, whichever is later.
8. **Differential of a fit at 34 weeks in Uganda?** Eclampsia, cerebral malaria, hypoglycaemia, epilepsy, meningitis, CVT/stroke, HIV-related CNS infection.
9. **What does HELLP stand for, and what are the thresholds?** Haemolysis (LDH ≥600), Elevated Liver enzymes (AST ≥70), Low Platelets (<100).
10. **What is the treatment target for BP?** 140–150/90–100 acutely (NICE target 135/85 for maintenance); avoid a precipitous fall.

On the ward round

1. **What would you do in the first 5 minutes for BP 170/115 with headache?** Repeat BP within 15 min, call for help, nifedipine IR 10 mg orally (or hydralazine 5 mg IV), $MgSO_4$ loading dose, IV access and bloods, catheter, fluid restriction, FHR, plan delivery.
2. **Which bloods do you send, and why each?** FBC (platelets, HELLP, anaesthesia), LFT and LDH (HELLP/liver), creatinine (kidney, $MgSO_4$ dosing), clotting/bedside clotting test (DIC, abruption), group & cross-match (PPH), glucose and mRDT (fit mimics).
3. **You find RR 12 and absent reflexes at the 4-hourly check. What do you do?** Withhold the dose, call for help, oxygen, calcium gluconate 1 g IV over 10 min if respiration is depressed or she stops breathing, ventilate, check urine output and creatinine.
4. **Why can't she have a spinal?** Platelets below 75–80 × 10^9/L risk a spinal/epidural haematoma; use general anaesthesia with care to blunt the intubation pressor response, or transfuse platelets.

5. **How do you tell HELLP from AFLP?** AFLP has **hypoglycaemia**, more jaundice, marked coagulopathy and often less hypertension; HELLP has haemolysis (LDH ≥600, schistocytes) and thrombocytopenia with milder coagulopathy. Both need delivery.
6. **When does $MgSO_4$ stop, and why keep watching BP after delivery?** 24 h after delivery or the last fit. BP peaks on days 3–6 and up to a third of eclamptic fits are postpartum.
7. **She is 30 weeks with severe PE and stable. What is the plan?** $MgSO_4$ (also fetal neuroprotection <32 weeks), BP control, steroids, deliver after 48 h if stable at a centre with NICU, or sooner for any maternal or fetal deterioration.
8. **What will you tell her before discharge?** Danger signs, BP check within 1 week, 6-week review, aspirin from 12 weeks next pregnancy, lifelong cardiovascular risk, avoid combined oestrogen contraception while hypertensive.

Mnemonics

> EXAM FAVOURITE
> **Severe features: the "7 P's"**: **P**ressure ≥160/110, **P**ain in the head (headache), **P**hotopsia (visual disturbance), **P**ain in the epigastrium/RUQ, **P**ulmonary oedema, **P**latelets <100, **P**ee (oliguria or rising creatinine).

- **$MgSO_4$ monitoring: "R-R-U"**: **R**espiratory rate ≥16, **R**eflexes present, **U**rine ≥30 mL/h. And **"C for Calcium, C for Crisis"**.
- **Toxicity ladder: "Knees, then Breathing, then Heart"**: reflexes lost (4–5 mmol/L) → respiratory depression (6–7.5) → cardiac arrest (>12).
- **Pritchard = "4 + 5 + 5 then 5 every 4"**: 4 g IV, 5 g IM each buttock, then 5 g IM every 4 h.
- **Acute antihypertensives: "No Hurry? Labetalol"** - **N**ifedipine, **H**ydralazine, **L**abetalol. **Methyldopa is "Maintenance only"**.
- **Drugs to avoid: "ACE, ARB, Atenolol, Ergometrine, Excess fluids"**.
- **Eclampsia delivery clock**: "**12 hours** for a fit, **24 hours** for severe, **37 weeks** for mild".

OSCE tips

- **BP measurement station**: explain, seat her or position left lateral at 45°, choose the correct cuff, palpate the systolic first, use **Korotkoff V**, repeat, interpret, and **always dip the urine**.
- **$MgSO_4$ preparation station**: state the strengths aloud (50% = 0.5 g/mL); make up 20% (8 mL of 50% + 12 mL diluent); give 5 g IM + 1 mL 2% lignocaine per buttock; state the monitoring and the antidote.
- **Counselling station**: explain pre-eclampsia in plain words ("the placenta is not getting enough blood and is releasing substances that affect your blood vessels"), why delivery is needed, danger signs, aspirin next time, and the long-term heart risk.
- **Eclampsia drill station**: call for help, left lateral, airway and oxygen, nothing in the mouth, $MgSO_4$ loading dose, glucose and mRDT, BP control, deliver within 12 h.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define pre-eclampsia (ISSHP 2021).	New hypertension (**≥140/90**) at or after **20 weeks** PLUS ≥1 of: **proteinuria**, **maternal organ dysfunction** (neuro, pulmonary oedema, haematological, hepatic, renal) or **uteroplacental dysfunction** (FGR, abnormal Doppler, stillbirth).
2	Thresholds for significant proteinuria?	Dipstick **≥2+** (1+ needs confirming), urine PCR **≥30 mg/mmol**, or **≥300 mg/24 h**.
3	What BP defines severe hypertension, and how fast must it be treated?	**≥160 systolic and/or ≥110 diastolic**, confirmed within 15 min. Treat within **30–60 minutes** to prevent intracerebral haemorrhage.
4	Pritchard $MgSO_4$ regimen (Uganda standard)?	**Loading**: 4 g of 20% IV over 5–10 min + 10 g of 50% IM (5 g each buttock with 1 mL 2% lignocaine) = 14 g. **Maintenance**: 5 g of 50% IM every 4 h, alternate buttocks, for **24 h after delivery or the last fit**.
5	Zuspan $MgSO_4$ regimen?	**4 g IV over 5–20 min**, then **1 g/hour IV infusion** for 24 h after delivery or last fit.
6	Three checks before every $MgSO_4$ maintenance dose?	1. **RR ≥16/min** 2. **Patellar reflexes present** 3. **Urine output ≥30 mL/h** (≥100 mL in 4 h)
7	Antidote for $MgSO_4$ toxicity and its dose?	**Calcium gluconate 1 g (10 mL of 10%) IV slowly over 10 minutes**, plus ventilation support; stop $MgSO_4$.
8	Serum Mg levels: therapeutic, loss of reflexes, respiratory depression, cardiac arrest?	Therapeutic **2–3.5 mmol/L**; reflexes lost **4–5**; respiratory depression **6–7.5**; cardiac arrest **>12 mmol/L**. Order: knees, breathing, heart.
9	How do you make 20% $MgSO_4$ for IV loading from 50% ampoules?	**8 mL of 50% (4 g) + 12 mL water for injection/normal saline = 20 mL of 20%**, given slowly over 5–10 min.
10	Hydralazine dose for severe hypertension?	**5 mg IV slowly** (over 2–5 min); recheck BP in 20 min; repeat 5–10 mg every 20–30 min to a **max of 20 mg IV** (WHO alternative: 12.5 mg IM every 2 h).
11	Oral nifedipine for acute severe hypertension?	**Nifedipine immediate-release 10 mg orally** (swallowed, not sublingual); repeat after 20–30 min if still ≥160/110; then nifedipine MR 20 mg 12-hourly.
12	IV labetalol regimen for severe hypertension?	**20 mg IV over 2 min**, then **40, 80, 80 mg every 10 min** as needed; **max 300 mg**. Avoid in asthma.
13	Why is methyldopa not used for acute severe hypertension, and why switch it after delivery?	**Slow onset** (4–6 h or longer). Postpartum it is linked to **depression**, so switch to nifedipine, amlodipine or enalapril within about 2 days.
14	HELLP syndrome: components and laboratory thresholds?	**H**aemolysis (LDH ≥600 IU/L, schistocytes), **E**levated **L**iver enzymes (AST ≥70 IU/L), **L**ow **P**latelets (<100 × 10^9/L).
15	Core pathophysiology of pre-eclampsia in one line?	Failed trophoblast remodelling of spiral arteries → placental ischaemia/oxidative stress → release of **sFlt-1 and soluble endoglin** (anti-angiogenic) → systemic **endothelial dysfunction**.

#	Question	Answer
16	Who gets aspirin for prevention, and at what dose?	One high-risk factor (previous PE, chronic HTN, CKD, SLE/APS, diabetes) or ≥2 moderate (primigravida, age ≥40, interval >10 y, BMI ≥35, family history, multiple pregnancy): **aspirin 75–150 mg at night from 12 weeks (before 16–20) until 36 weeks**.
17	WHO calcium supplementation for pre-eclampsia prevention?	**1.5–2.0 g elemental calcium daily** (divided doses) from 20 weeks to delivery in populations with low dietary calcium; take it separately from iron.
18	Timing of delivery: eclampsia, severe PE ≥34 wk, PE without severe features?	**Eclampsia**: within 12 h of stabilisation. **Severe PE ≥34 weeks**: once stable (within 24 h). **PE without severe features**: at **37+0 weeks** (HYPITAT).
19	Uterotonic of choice for the third stage in pre-eclampsia, and which to avoid?	**Oxytocin 10 IU IM/IV**. **Avoid ergometrine/Syntometrine**: it causes vasoconstriction, hypertensive crisis and stroke.
20	Maximum IV/oral fluid intake in severe pre-eclampsia, and why?	About **80 mL/h (1 mL/kg/h)**. Leaky capillaries and low oncotic pressure mean extra fluid causes **pulmonary oedema**; oliguria is from renal vasospasm, not dehydration.
21	First 5 minutes: a pregnant woman at 34 weeks with BP 168/114 and severe headache. What are your immediate actions?	Call for help; **repeat BP within 15 min**; give **nifedipine IR 10 mg orally** (or **hydralazine 5 mg IV**) to treat within 30–60 min; **$MgSO_4$ loading dose** (Pritchard); IV access + bloods; **glucose and mRDT**; **catheter** with hourly urine; restrict fluids to **80 mL/h**; check FHR; plan delivery.
22	How do you perform and interpret the WHO bedside clotting test?	Put **2 mL of venous blood** in a small plain dry glass tube, hold it in your closed fist (keep it warm), and tip it gently every minute from 4 minutes. **No clot by 7 minutes, or a soft clot that breaks up = coagulopathy** (DIC, low platelets).
23	PET bloods: platelets 64, AST 186, LDH 910, creatinine 98, glucose 5.2. Interpret.	**Complete HELLP** (platelets <100, AST ≥70, LDH ≥600) with **renal involvement** (creatinine >90 µmol/L). Normal glucose argues against AFLP. **Deliver**; platelets <75–80 → no spinal; <50 before CS → transfuse platelets; watch urine output before each $MgSO_4$ dose.
24	On $MgSO_4$: RR 12/min, patellar reflexes absent, urine 60 mL in 4 h. What do you do?	**Magnesium toxicity**: **withhold** the dose, call for help, oxygen/airway; **calcium gluconate 1 g (10 mL of 10%) IV over 10 min** if breathing is depressed or stops (ventilate if needed); check creatinine ± serum Mg. Restart only when reflexes return, RR ≥16 and urine ≥30 mL/h.
25	Dipstick 3+ protein, BP 150/100, no symptoms, normal bloods at 35 weeks: severe or not? What is the plan?	**Pre-eclampsia WITHOUT severe features**: the amount of proteinuria does not grade severity. Admit, BP 4-hourly, bloods twice weekly, fetal surveillance, oral antihypertensive if persistently ≥140/90, and **deliver at 37+0 weeks** (sooner if severe features appear).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 22-year-old primigravida at 36 weeks presents to Mengo Hospital with a severe frontal headache. BP is 168/112 mmHg and 172/114 mmHg 15 minutes later. Urine protein is 2+. She has no IV access yet. What is the MOST appropriate immediate antihypertensive?

- A Methyldopa 500 mg orally
- B Enalapril 10 mg orally
- C Furosemide 40 mg IV
- D Nifedipine immediate-release 10 mg orally
- E Atenolol 50 mg orally

2. A 27-year-old G2P1 at 34 weeks with severe pre-eclampsia received a Pritchard loading dose of $MgSO_4$ 4 hours ago. Before the next dose, her RR is 18/min, patellar reflexes are present, and urine output over the last 4 hours is 60 mL. What is the MOST appropriate action?

- A Give the 5 g IM dose as scheduled
- B Withhold the maintenance dose, review renal function and resume when urine output improves
- C Give calcium gluconate 1 g IV
- D Give a 2 L normal saline bolus and then the full dose
- E Stop $MgSO_4$ permanently and switch to diazepam

3. A 24-year-old on $MgSO_4$ Pritchard maintenance for eclampsia becomes drowsy 2 hours after an IM dose. RR is 8/min and patellar reflexes are absent. SpO_2 is 88%. After stopping $MgSO_4$ and supporting ventilation, what should be given?

- A Naloxone 0.4 mg IV
- B Diazepam 10 mg IV
- C Sodium bicarbonate 50 mmol IV
- D Hydralazine 5 mg IV
- E Calcium gluconate 1 g (10 mL of 10%) IV over 10 minutes

4. A 19-year-old primigravida at 38 weeks is brought to HC IV after a generalised seizure at home. On arrival she is post-ictal, BP 160/105 mmHg, proteinuria 3+. While you are preparing drugs she has another seizure. What is your FIRST action?

- A Protect her from injury, and after the fit place her in the left lateral position and clear the airway
- B Insert a mouth gag to protect her tongue
- C Arrange immediate caesarean section
- D Give hydralazine 10 mg IV
- E Do a lumbar puncture to exclude meningitis

5. A 30-year-old G3P2 at 33 weeks presents with epigastric pain and vomiting, which the clinician attributed to gastritis. BP is 148/96 mmHg, urine protein 1+. Hb 9.8 g/dL, platelets 68×10^9/L, AST 210 IU/L, LDH 820 IU/L. What is the MOST likely diagnosis?

- A Acute fatty liver of pregnancy
- B Viral hepatitis
- C HELLP syndrome
- D Peptic ulcer disease
- E Acute cholecystitis

6. A 28-year-old at 37+2 weeks has pre-eclampsia without severe features. BP is 146/94 mmHg on nifedipine MR, bloods are normal and the fetus is well grown. Cervix: Bishop score 3. What is the MOST appropriate plan?

A Expectant management until 40 weeks
B Induction of labour with cervical ripening (misoprostol or Foley catheter)
C Elective caesarean section today
D Discharge home with weekly review
E Start $MgSO_4$ and deliver by caesarean within 12 hours

7. A 35-year-old para 4 delivers vaginally at a regional referral hospital. She has had pre-eclampsia with severe features managed with $MgSO_4$ and hydralazine. The midwife prepares the third-stage uterotonic. Which drug should be AVOIDED?

A Oxytocin 10 IU IM
B Carbetocin 100 mcg IM
C Misoprostol 600 mcg orally
D Ergometrine 0.5 mg IM
E Tranexamic acid 1 g IV if bleeding

8. A 23-year-old at 30 weeks with severe pre-eclampsia is on $MgSO_4$ at a HC IV. BP is controlled at 150/95 on nifedipine. Platelets 160, creatinine 65 µmol/L, AST 30. CTG is reassuring. There is no NICU at the HC IV. What is the MOST appropriate next step? Advanced

A Give dexamethasone and arrange in-utero transfer to a referral hospital with NICU
B Immediate caesarean section at the HC IV
C Stop $MgSO_4$ and discharge on methyldopa
D Induce labour with oxytocin at the HC IV
E Start expectant management at the HC IV until 37 weeks

9. A 26-year-old at 35 weeks with severe pre-eclampsia is oliguric (15 mL/h for 2 hours) after delivery by caesarean section. She has received 3 L of IV fluid in 6 hours. She is now breathless, RR 32/min, SpO_2 86% with bilateral basal crackles. What is the MOST appropriate immediate management? Advanced

A Give a further 500 mL fluid challenge for oliguria
B Give hydralazine 20 mg IV bolus
C Give calcium gluconate 1 g IV
D Start dopamine infusion
E Sit her up, give oxygen and furosemide 40 mg IV, and stop IV fluids

10. A 31-year-old G3P2 books at 10 weeks at an urban HC IV. She had eclampsia at 32 weeks in her last pregnancy. BP today is 118/72 mmHg. What is the MOST appropriate preventive intervention?

A Methyldopa 250 mg 8-hourly from booking
B $MgSO_4$ prophylaxis from 28 weeks
C Aspirin 75–150 mg at night from 12 weeks until 36 weeks, plus calcium supplementation
D Vitamin C and E supplements
E Bed rest from 20 weeks

11. A 20-year-old at 36 weeks has a fit at a rural HC III that has $MgSO_4$ but no theatre or IV antihypertensives. BP after the fit is 172/112 mmHg. The nearest HC IV with theatre is 40 km away. What is the BEST course of action?

A Give the $MgSO_4$ loading dose and oral nifedipine 10 mg, then refer urgently with an escort and the next dose

B Refer immediately without treatment to save time

C Give diazepam 10 mg IV and observe overnight

D Give $MgSO_4$ and keep her at the HC III to deliver vaginally

E Give oxytocin to induce labour before referral

12. A 29-year-old with severe pre-eclampsia at 37 weeks needs an emergency caesarean section for fetal distress. Platelets are 58×10^9/L. BP is 150/100 on treatment. Which statement about anaesthesia and preparation is MOST appropriate? Advanced

A Spinal anaesthesia is safe at any platelet count

B Give ergometrine at delivery of the placenta

C Stop $MgSO_4$ before surgery because it interacts with anaesthesia

D General anaesthesia is preferred and the intubation pressor response must be blunted; arrange platelets for transfusion

E Postpone surgery until platelets exceed 100

13. A 24-year-old on day 4 after a normal delivery presents to Mengo casualty with a severe headache and blurred vision. She had mildly raised BP in labour and was discharged on day 1. BP is 164/110 mmHg, urine protein 2+. What is the MOST appropriate management?

A Reassure her; pre-eclampsia resolves after delivery

B Treat as postpartum pre-eclampsia with severe features: acute antihypertensive, $MgSO_4$ and admission

C Give paracetamol and review in the outpatient clinic in a week

D Do a lumbar puncture first

E Start oral methyldopa and discharge

14. A 26-year-old at 32 weeks presents with a generalised seizure, temperature 39.4°C, BP 118/74 mmHg and no proteinuria. She lives in Gulu and has not slept under a net. Blood glucose is 2.4 mmol/L. What is the MOST likely diagnosis?

A Eclampsia

B Epilepsy

C Postpartum pre-eclampsia

D Cerebral venous sinus thrombosis

E Severe (cerebral) malaria with hypoglycaemia

15. A 32-year-old with chronic hypertension on enalapril 10 mg daily attends her first ANC visit at 8 weeks. BP is 136/86 mmHg. What is the MOST appropriate change to her medication?

A Continue enalapril and add aspirin

B Stop all antihypertensives until 20 weeks

C Stop enalapril and switch to methyldopa or nifedipine MR, and start aspirin at 12 weeks

D Switch to losartan

E Add furosemide

Answers & explanations

1. D · Nifedipine immediate-release 10 mg orally

Severe hypertension (≥160/110) must be lowered within 30–60 minutes. **Oral immediate-release nifedipine 10 mg** works in 10–20 minutes, needs no IV line, and is widely available in Uganda. **Methyldopa** has a slow onset (hours) and is for maintenance only. **Furosemide** reduces an already depleted intravascular volume and is only for pulmonary oedema. **Enalapril** (ACE inhibitor) is contraindicated antenatally (fetal renal damage). **Atenolol** is slow, not an acute agent, and is associated with FGR.

2. B · Withhold the maintenance dose, review renal function and resume when urine output improves

Urine output is **<100 mL in 4 hours**. Magnesium is renally excreted, so oliguria leads to accumulation and toxicity. **Withhold** (or reduce) the dose, check creatinine, and resume when urine output is ≥30 mL/h. Giving the dose as scheduled risks toxicity. Calcium gluconate is for **toxicity** (respiratory depression/arrest), which is not present. A **2 L bolus** risks pulmonary oedema; oliguria in PE is from renal vasospasm. Diazepam is inferior to $MgSO_4$ and is used only when $MgSO_4$ is unavailable.

3. E · Calcium gluconate 1 g (10 mL of 10%) IV over 10 minutes

Absent reflexes and respiratory depression indicate **magnesium toxicity**; the antidote is **calcium gluconate 1 g IV slowly over 10 min**, with bag-mask ventilation. **Naloxone** reverses opioids, not magnesium. **Diazepam** would worsen respiratory depression. **Hydralazine** lowers BP and has no role here. **Bicarbonate** does not reverse Mg-induced neuromuscular block.

4. A · Protect her from injury, and after the fit place her in the left lateral position and clear the airway

Eclampsia first aid is **ABC**: protect from injury (do not restrain), then left lateral position, suction, oxygen, then $MgSO_4$. **Nothing should be put in the mouth** (it causes injury). **Caesarean** is only after the mother is stabilised; an eclamptic woman is too unstable for immediate theatre. BP control matters but comes **after** airway and $MgSO_4$. **LP** is not a first step and is dangerous with raised intracranial pressure.

5. C · HELLP syndrome

Haemolysis (LDH >600, falling Hb), elevated AST (>70) and low platelets (<100) with hypertension = **HELLP**; BP and proteinuria may be only modestly raised. **AFLP** features hypoglycaemia, jaundice and coagulopathy, with less haemolysis and thrombocytopenia. **Viral hepatitis** gives much higher transaminases and no haemolysis/thrombocytopenia pattern. **PUD** and **cholecystitis** do not explain haemolysis and thrombocytopenia; calling this 'gastritis' is a classic, dangerous error.

6. B · Induction of labour with cervical ripening (misoprostol or Foley catheter)

At **≥37 weeks**, pre-eclampsia should be delivered (HYPITAT trial: induction reduces maternal complications without raising the caesarean rate). With an unfavourable cervix, **ripen with misoprostol 25 mcg or a Foley catheter**, then oxytocin. **Waiting** exposes her to the risk of progression. **Caesarean** is for obstetric indications, not pre-eclampsia per se. **Discharge** is unsafe at term with PE in Uganda. The **12-hour** delivery window applies to eclampsia.

7. D · Ergometrine 0.5 mg IM

Ergometrine (and Syntometrine) causes generalised vasoconstriction and can trigger a **hypertensive crisis, stroke or eclampsia**; it is contraindicated in hypertensive disorders. **Oxytocin** is the drug of choice. **Carbetocin** and **misoprostol** do not raise BP significantly and are acceptable alternatives. **TXA** is used for PPH treatment and is not contraindicated.

8. A · Give dexamethasone and arrange in-utero transfer to a referral hospital with NICU

She is stable, preterm and her fetus would need neonatal intensive care. The best plan is **corticosteroids** (dexamethasone 6 mg IM 12-hourly × 4) and **in-utero transfer** ('the best incubator is the uterus during transport') to a centre with NICU and HDU, where expectant management or delivery after steroids can occur. **Immediate CS** is not indicated while stable. **Discharge** is unsafe with severe disease. **Induction at the HC IV** gives a 30-week baby without NICU. **Expectant management to 37 weeks** is inappropriate for severe disease.

9. E · Sit her up, give oxygen and furosemide 40 mg IV, and stop IV fluids

This is **pulmonary oedema** from fluid overload on top of capillary leak. Treatment: **sit up, oxygen, furosemide 20–40 mg IV, stop fluids**, strict input/output and senior/anaesthetic review. **More fluid** worsens it; oliguria in PE is renal vasospasm. **Calcium gluconate** treats magnesium toxicity, which is not the problem. **Dopamine** has no proven benefit. A **20 mg hydralazine bolus** risks precipitous hypotension; lower BP carefully if severe.

10. C · Aspirin 75–150 mg at night from 12 weeks until 36 weeks, plus calcium supplementation

Previous pre-eclampsia/eclampsia is a **high-risk factor**: give **low-dose aspirin from 12 weeks (before 16–20 weeks) until 36 weeks**, and **calcium** where dietary intake is low. **Methyldopa** treats hypertension; she is normotensive. **$MgSO_4$** is not a long-term prophylactic. **Vitamins C and E** were shown to be ineffective in large trials. **Bed rest** has no proven benefit and increases VTE risk.

11. A · Give the $MgSO_4$ loading dose and oral nifedipine 10 mg, then refer urgently with an escort and the next dose

At HC III the job is to **stabilise and refer**: $MgSO_4$ **loading dose** (IM 10 g ± IV 4 g), a **first dose of antihypertensive** (nifedipine 10 mg orally once she can swallow safely), then urgent referral with a health worker, notes and the next $MgSO_4$ dose. **Diazepam** is inferior and she needs delivery within 12 h. **Referral without treatment** risks further fits and stroke on the way. **Keeping her** at HC III risks death without surgical or HDU backup. **Induction** before stabilisation and at a facility without theatre is unsafe.

12. D · General anaesthesia is preferred and the intubation pressor response must be blunted; arrange platelets for transfusion

Neuraxial anaesthesia is generally considered safe with platelets **≥75–80 × 10^9/L**; at 58 there is a risk of spinal haematoma, so **general anaesthesia** is used, with measures (e.g. short-acting opioid) to **blunt the hypertensive response to laryngoscopy** (a risk for stroke). **Platelet transfusion** is advised when <50 before CS. $MgSO_4$ should be **continued** (anaesthetists adjust muscle relaxant doses). **Ergometrine** is contraindicated. **Postponing** surgery for fetal distress is wrong.

13. B · Treat as postpartum pre-eclampsia with severe features: acute antihypertensive, $MgSO_4$ and admission

Pre-eclampsia and eclampsia can occur **up to 6 weeks postpartum**, with BP peaking on **days 3–6**. Severe BP with neurological symptoms = **severe features**: control BP (nifedipine IR or hydralazine), give **$MgSO_4$** for 24 h, admit, and check bloods. **Reassurance** or **paracetamol and review** risks eclampsia or stroke. **LP** is not the first step (think CVT if atypical: CT). **Methyldopa** is slow and is avoided postpartum (depression).

14. E · Severe (cerebral) malaria with hypoglycaemia

Fever, normal BP, **no proteinuria** and **hypoglycaemia** in a high-transmission area point to **cerebral malaria** (pregnancy increases susceptibility and hypoglycaemia). Treat with IV artesunate and IV dextrose. **Eclampsia** is less likely with normal BP and no proteinuria (though it can occur with modest BP; if unsure, giving $MgSO_4$ is reasonable while testing). **Epilepsy** does not explain fever and hypoglycaemia. **CVT** is usually postpartum with focal signs. She has not delivered, so it is not **postpartum**.

15. C · Stop enalapril and switch to methyldopa or nifedipine MR, and start aspirin at 12 weeks

ACE inhibitors (and **ARBs** like losartan) are fetotoxic in the second and third trimesters (renal failure, oligohydramnios, skull defects), so switch to a pregnancy-safe agent (**methyldopa, nifedipine or labetalol**). Chronic hypertension is a **high-risk factor**, so give **aspirin from 12 weeks**. **Stopping all drugs** risks uncontrolled BP. **Diuretics** reduce plasma volume and are avoided.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Eclampsia on arrival at a HC IV

A 17-year-old primigravida at about 36 weeks (unsure of dates, 2 ANC visits) is carried into the maternity unit of a HC IV in Wakiso by her mother. She had a fit at home 1 hour ago and another on the boda boda. On arrival she is drowsy, BP 176/118 mmHg, pulse 110, RR 24, SpO_2 93% on air. Her face and hands are swollen.

1. What are your immediate priorities in the first 5 minutes?

> Call for help (midwife, MO, anaesthetist). **ABC**: left lateral position, clear airway and suction, **oxygen 4–6 L/min**, protect from injury, do not put anything in the mouth. IV access with a 16–18 G cannula, draw bloods. **$MgSO_4$ Pritchard loading dose now** (4 g of 20% IV over 5–10 min + 5 g of 50% IM into each buttock with 1 mL 2% lignocaine). Check **blood glucose** and an **mRDT**.

2. What focused history do you take from her mother while treating?

> Headache, blurred vision, epigastric pain before the fits (supports eclampsia). Fever, rigors, neck stiffness (malaria, meningitis). Known epilepsy, previous fits, HIV status, drugs. LNMP and ANC records (gestation, previous BP readings). Fetal movements, vaginal bleeding or abdominal pain (abruption), leaking liquor, contractions (labour). Time of the last meal (aspiration risk).

3. BP remains 172/114 after the loading dose. Urine dipstick 3+ protein. Glucose 5.8, mRDT negative. What next?

> Treat severe BP within 30–60 min: **hydralazine 5 mg IV slowly**, repeat 5–10 mg every 20–30 min to max 20 mg, or **nifedipine IR 10 mg orally** once she is awake enough to swallow. Target 140–150/90–100. **Foley catheter** with hourly urine output; **restrict fluids** to 80 mL/h. Send FBC, LFT, creatinine, clotting time, group & cross-match. Assess the fetus (FHR; expect transient bradycardia after a fit). Monitor BP, RR, reflexes, SpO_2 and GCS every 15–30 min.

4. The fetus is cephalic, FHR 150 and regular. Cervix 2 cm, posterior, firm (Bishop 3). She is now awake. How and when should she be delivered?

> Deliver **within 12 hours** of the fit. With a Bishop score of 3 and a primigravida, vaginal delivery within 12 hours is unlikely; many clinicians would do a **caesarean section** once stable (spinal anaesthesia if platelets ≥75–80). An alternative, if the unit has capacity and she remains stable, is rapid ripening (Foley catheter or misoprostol) with a plan to proceed to CS if not in established labour. Oxytocin 10 IU for the third stage, **no ergometrine**. Inform the neonatal team (possible preterm baby, $MgSO_4$ hypotonia).

5. Four hours after delivery, her RR is 12/min and patellar reflexes are absent. What do you do?

> This is **$MgSO_4$ toxicity**. **Withhold** further $MgSO_4$, support breathing (oxygen, bag-mask if needed), give **calcium gluconate 1 g (10 mL of 10%) IV over 10 min**. Check urine output and creatinine (renal impairment is the usual cause). When reflexes return and RR ≥16, resume at a reduced dose if still needed. Document and review the monitoring chart.

6. What postpartum care and counselling does she need before discharge?

> Continue $MgSO_4$ for **24 h after delivery or last fit**. BP 4-hourly; continue oral antihypertensives (nifedipine MR or enalapril; avoid methyldopa postpartum) and wean as BP falls. Repeat FBC, LFT and creatinine. Watch for pulmonary oedema. Discharge when BP <150/100 with symptoms resolved; **BP review within 1 week** and at 6 weeks. Counsel: danger signs, **aspirin from 12 weeks** in the next pregnancy, early ANC, recurrence risk, **long-term cardiovascular risk**. Contraception: avoid combined oestrogen methods while hypertensive; a progestogen implant or IUD is ideal for a 17-year-old.

> **Take-away:** In eclampsia: ABC, $MgSO_4$, control BP, check glucose and malaria, restrict fluids, deliver within 12 hours, and watch the RR-Reflex-Urine triad, with calcium gluconate at the bedside.

Case 2: Epigastric pain at 33 weeks

A 30-year-old G3P2 at 33 weeks presents to Mengo Hospital with epigastric pain and vomiting for 12 hours. She was given omeprazole at a clinic yesterday. BP 152/98 mmHg, pulse 96, afebrile. Urine protein 1+. She has mild RUQ tenderness.

1. What is your working differential diagnosis?

> **HELLP syndrome/pre-eclampsia with liver involvement** (top of the list). Others: acute fatty liver of pregnancy, viral hepatitis, severe malaria with haemolysis, cholecystitis, pancreatitis, peptic ulcer, pyelonephritis, placental abruption (pain, but usually uterine).

2. Which investigations do you order, and why?

> **FBC** (platelets, Hb), **blood film** (schistocytes), **LDH**, **AST/ALT, bilirubin**, **creatinine/urea**, **glucose** (low in AFLP), **clotting** (PT/APTT/fibrinogen or bedside clotting time), **mRDT**, group & cross-match, urine PCR, **ultrasound** (fetal growth, liquor, placenta, liver haematoma), CTG.

3. Results: Hb 9.5 g/dL, platelets 72 × 10^9/L, AST 185 IU/L, LDH 790 IU/L, creatinine 88 µmol/L, glucose 5.2 mmol/L, schistocytes on film. Interpret.

> **HELLP syndrome**: haemolysis (LDH ≥600, schistocytes, falling Hb), elevated liver enzymes (AST ≥70), low platelets (<100). Normal glucose argues against AFLP. This is **pre-eclampsia with severe features** regardless of BP or proteinuria.

4. Outline your management.

> Admit to HDU. **$MgSO_4$** (Pritchard). Control BP if ≥160/110 (nifedipine/hydralazine). Fluid restriction 80 mL/h with catheter. **Dexamethasone 6 mg IM 12-hourly × 4** for fetal lungs (33 weeks). **Deliver** once stable, ideally after steroids (24–48 h) if the mother and fetus remain stable, **immediately** if DIC, abruption, liver haematoma or fetal compromise. Crossmatch blood; arrange platelets (transfuse if <50 before CS) and FFP if coagulopathic. Inform anaesthetist (platelets 72: GA may be required) and NICU. Repeat labs 6–12-hourly.

5. Twelve hours later she complains of sudden right shoulder-tip pain and becomes pale, pulse 130, BP 90/60. What do you suspect and what do you do?

> **Subcapsular liver haematoma with rupture** → haemorrhagic shock. Resuscitate (2 large-bore cannulas, warmed fluids, O-negative/crossmatched blood, activate massive transfusion), urgent ultrasound (free fluid), call the **general surgeon** and obstetrician; **laparotomy** with delivery and liver packing. Correct coagulopathy. This is a life-threatening complication with high mortality.

> **Take-away:** Epigastric pain plus hypertension in late pregnancy is HELLP until proven otherwise; check platelets, LFT and LDH before giving antacids.

Case 3: New hypertension at a routine ANC visit

A 25-year-old primigravida at 29 weeks attends routine ANC at a HC III near Mukono. BP is 146/94 mmHg, and 144/92 mmHg after 4 hours of rest. She has no symptoms. Urine dipstick: trace protein. Fundal height 27 cm. She lives 25 km from the nearest hospital.

1. What is the diagnosis at this point, and what features must you look for?

> **Gestational hypertension** (new BP ≥140/90 after 20 weeks, no significant proteinuria). Look for features that would make it **pre-eclampsia**: proteinuria (confirm with PCR), headache, visual symptoms, epigastric pain, clonus, low platelets, raised creatinine/transaminases, and **FGR** (fundal height 27 cm at 29 weeks is within range but needs a scan).

2. What should be done at the HC III?

> Refer (non-emergency, same day or within 24 h) to a HC IV/hospital for bloods, ultrasound and a treatment plan, because she lives far away and needs monitoring. Give health education on **danger signs** (headache, blurred vision, epigastric pain, fits, reduced fetal movements, bleeding). Ensure she is on iron/folate; calcium if possible.

3. At the hospital: platelets 210, creatinine 60, AST 25, PCR 45 mg/mmol, EFW 50th centile, normal liquor. BP 148/96. How do you classify it and manage it?

> **Pre-eclampsia without severe features** (proteinuria PCR ≥30). Admit (distance, reliability of follow-up). Start **nifedipine MR 20 mg 12-hourly or methyldopa 250 mg 8-hourly**, target about 135/85. BP 4-hourly, daily urine, bloods twice weekly, fetal movement chart, growth scan every 2 weeks. Plan **delivery at 37 weeks** unless severe features develop. Consider steroids if delivery before 34 weeks becomes likely.

4. At 33 weeks she develops BP 166/112 and a headache not relieved by paracetamol. Platelets 130. What is your plan?

> Now **severe features**. Acute BP control (nifedipine IR 10 mg or hydralazine 5 mg IV), **$MgSO_4$**, fluid restriction and catheter, **dexamethasone 6 mg IM 12-hourly × 4**, inform NICU. At 33 weeks with neurological symptoms, deliver once stabilised (after steroids if she stabilises quickly and symptoms settle; immediately if BP is uncontrollable, symptoms persist or fetal distress).

5. What long-term advice will you give her at discharge?

> BP check within 1 week, then at 6 weeks; if still hypertensive at 12 weeks, investigate for chronic hypertension/renal disease. Next pregnancy: early booking, **aspirin 75–150 mg from 12 weeks to 36 weeks**, calcium. Recurrence risk about 15–25%. Pre-eclampsia doubles her lifetime risk of **hypertension, heart disease and stroke**, so she needs annual BP checks and a healthy lifestyle.

Take-away: Gestational hypertension must be watched for progression; severe features can appear suddenly, and the plan escalates from monitoring to $MgSO_4$, steroids and delivery.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.